

6 Turbulent flows: model equations and solution methodology

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I INTRODUCTION

Most of the practical aerodynamic and hydrodynamic fluid flows of interest are turbulent. Attempts to understand the physics of such flows, as well as efforts to predict and control them, have been ongoing for 100 years. These efforts constitute the so-called 'turbulence problem', whose complete solution still eludes scientists and engineers. In the earlier part of this century, both theoretical and experimental efforts were made by many scientists; however, their understanding of the problem was incomplete, and the analyses constrained by the techniques available at the time.

The dawn of the supercomputer age brought hope that scientists would be

able to compute directly such flows from the Navier–Stokes equations and ultimately solve the turbulence problem. Unfortunately, even with the advent of the new generation of vector and parallel processors, the direct numerical simulation (DNS) of complex turbulent flows from the Navier–Stokes equations will probably not be possible for the foreseeable future. Because the problem is the inability to resolve all the component scales within the flow, an obvious approximation is to model the smaller scales and only simulate the larger scales. Such an alternative has been found in the large-eddy simulation (LES) technique. The Navier–Stokes equations are filtered so that the flow variables are split into resolvable and subgrid-scale motions. Suitable models are required for the subgrid-scale motions. The significant strides made in this area have been outlined in Chapter 5.

Although the simulation techniques mentioned above may ultimately become common engineering tools sometime in the next century, a void currently exists for computing complex engineering turbulent flows. The LES technique models only the unresolvable scales within the flow; the next level models the entire flow through suitable averaged quantities for both the mean and turbulent motion. This averaging procedure, known as Reynolds averaging, decomposes the motion into both mean and fluctuating components. Although the procedure eliminates the need to completely resolve the turbulent motion, its drawback is that unknown higher-order correlations appear in both the mean and turbulent equations. The need to model these correlations is the ‘closure problem’ associated with this technique. Nevertheless, the engineering tool of choice in computing turbulent flows is the Reynolds-averaged Navier–Stokes equations for the mean motion, coupled with a closure scheme for the unknown turbulent Reynolds stresses. From a physical standpoint, the task at hand is to somehow characterize the turbulence. One obvious characterization is by adequately describing the evolution of representative turbulent velocity and length scales, an idea that originated over 50 years ago (Kolmogorov, 1942). Normally, the velocity scale is characterized by the turbulent Reynolds stresses or turbulent kinetic energy, and the length scale is characterized by coupling the turbulent energy dissipation rate with this velocity scale. Although this specific representation is not unique, the physical cornerstone behind the development of turbulent closure models is this ability to model correctly the characteristic scales associated with the turbulent flow.

The focus of this chapter is to outline the development of the turbulent closure schemes and to discuss their implementation into numerical algorithms. An unambiguous evaluation of the performance of the various second-moment or two-equation models is difficult, if not impossible, to provide. The main problems are how to isolate the behaviour of the individual correlations and how to perform the calculations so that the results are not dependent on the numerical test bed. Alternative attempts at model evaluation require very detailed experiments or simulations. The experiments require hard to measure correlations which, even if obtained, may not be sufficiently accurate, and the

direct numerical simulations are of limited use because the Reynolds numbers are low and, as such, are inconsistent with the evaluation of the high-Reynolds-number models. Thus, the number of such experimental test flows is rather small. Nevertheless, attempts have been made to evaluate models, and these studies are referenced throughout the chapter. The reader is encouraged to review this material before embarking on any extensive calculations of turbulent flows. The intent of this chapter is to provide the reader with a knowledge of the current trends in turbulent modelling research, as well as an up-to-date outline of the capabilities available to the researcher in solving turbulent flow problems. Additional topics that relate to multiple-species turbulent flows or reacting turbulent flows are outside the scope of this chapter.

In the first part of the chapter, the problem of solving turbulent flow fields, is formulated mathematically. The mean and turbulent equations derived are applicable to both incompressible and compressible flows. The turbulent closure problem is identified, and a hierarchy of closure models is discussed applicable to high-Reynolds number flows. Also discussed is the wall-function approach, and the alternative near-wall modifications to the high-Reynolds-number forms of the models.

In the second part of the chapter, the numerical aspects of solving the Reynolds-averaged Navier-Stokes equations and the turbulent closure equations are presented. Lower order models such as the algebraic and half-equation models simply impact the Navier-Stokes equations through a variable viscosity, and the half-equation model only adds an ordinary differential equation as an extension to the algebraic model. Thus, the viscous Navier-Stokes solution techniques, described in the previous chapters, suffice as solution methodologies for these types of turbulent closure schemes.

II INCOMPRESSIBLE TURBULENT MODELLING

In this section, the appropriate governing equations for the solution of incompressible turbulent flows are developed. The basis of the formulation is the continuity equation (mass conservation) and the Navier-Stokes equations given by

$$\frac{\partial u_j}{\partial x_j} = 0 \quad (1)$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{1}{\rho} \frac{\partial \sigma_{ij}}{\partial x_j} \quad (2)$$

Above, σ_{ij} is the deviatoric part of the viscous stress tensor and is given by the newtonian relationship

$$\sigma_{ij} = 2\mu S_{ij}. \quad (2a)$$

where ρ is the density, u_i is the fluid velocity, p is the pressure, μ is the molecular viscosity, and $S_{ij} = \frac{1}{2}(\partial u_i / \partial x_j + \partial u_j / \partial x_i)$ is the strain rate tensor. For the present purposes, the viscosity μ is assumed to be constant throughout the discussion on incompressible flows. Throughout this chapter, cartesian tensor notation, along with the Einstein summation convention, is adopted in the presentation of the governing equations and models. Readers not familiar with this notation will need to review one of the many texts that contain introductory tensor analysis (e.g. Aris, 1962; Kundu, 1990).

In a Reynolds decomposition, any flow variable can be decomposed into mean and fluctuating components as

$$f = \bar{f} + f', \quad (3)$$

where the mean variable $\bar{f}$ can be extracted if a statistically steady or a statistically homogeneous turbulence is assumed. In the former case, a long-time average of f is taken; in the latter case, a volume average is taken. If the flow is neither *stationary* nor *homogeneous*, and the average characteristics *vary with time or space*, then an *ensemble mean* over all realizations (samples) is required. For example, if the turbulence is stationary, then the appropriate average is

$$\bar{f}(\mathbf{x}) = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_{t_0}^{t_0 + \tau} f(\mathbf{x}, t) dt, \quad (3a)$$

whereas for a homogeneous turbulence the formula is similar but with the average taken over a large spatial volume. Whichever procedure is used, the average of a fluctuating quantity is zero ($\bar{f}' = 0$), and the average of the product of two quantities is the equality $\overline{f'g'} = \bar{f}'\bar{g}'$.

A Reynolds-averaged Navier–Stokes equations

If the procedure just outlined is applied to the continuity and Navier–Stokes equations for a constant property fluid, then

$$\frac{\partial \bar{u}_j}{\partial x_j} = 0 \quad (4)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{1}{\rho} \frac{\partial \bar{\sigma}_{ij}}{\partial x_j} - \frac{\partial \overline{u'_i u'_j}}{\partial x_j} \quad (5)$$

Equation (5) is the Reynolds-averaged Navier–Stokes equation (RANS). It has the same form as the instantaneous equation (2), with the exception of the second-moment or correlation tensor $\tau_{ij} \equiv \overline{u'_i u'_j}$. The Reynolds stress tensor is then given by $-\rho \tau_{ij}$, which is the basis for reference to Reynolds stress modelling. For consistency throughout this chapter and with most of the cited

literature, the quantity τ_{ij} is preferred; however, the terminology Reynolds stress or Reynolds stress modelling will be used freely because of the implied constant-density formulation throughout the incompressible discussion.

As is readily seen, (5) is not closed because the turbulent correlation τ_{ij} is not known. This closure problem is well known in association with the solution of the RANS. Transport equations for τ_{ij} can be derived by taking the second moment of the fluctuating momentum equation. These equations then contain triple moments, which also must be determined. All higher-order moment equations generate still unknown correlations, and the process must be truncated at some level. This discussion leads to the topic of phenomenological turbulence modelling in which a variety of closure models, which range in complexity from very simple algebraic models to fully coupled sets of partial differential equations that represent the higher-order moments, are used to model τ_{ij} .

B Turbulent correlation closure models

In this section, the hierarchy of closure models for the second moment or Reynolds stress tensor will be discussed. These closure models are generally presented in order of increasing complexity, starting with the algebraic or zero-equation models, and ending with the full Reynolds stress model. Although this development makes sense from the standpoint of introducing models of increasing difficulty, it masks common theoretical threads that underlie the models. For this reason, this chapter starts with the full Reynolds stress closure, which is easily derivable from the fluctuating momentum equation. This closure has always been viewed as the most difficult to implement because it leads to six partial differential equations for the independent stress components plus a differential scale equation that must be solved. However, with the exception of the algebraic or zero-equation models to be discussed, the other closure models are simplifications, or truncations, of the full Reynolds stress closure. This linkage needs to be more explicitly presented to show both the theoretical connection and the commonality of the assumptions that underlie most of the models.

1 Reynolds stress models

A straightforward approach to finding the appropriate closure for τ_{ij} is to simply form the second-moment equation and attempt to find models for the higher-order correlations in the equation. The starting point for developing the second moment or the Reynolds stress transport equations is the fluctuating momentum equation, which is obtained by subtracting the mean momentum equation (5) from (2) and using the Reynolds decomposition (3). If we write the

resulting fluctuating momentum equation for u'_i

$$\mathcal{L}u'_i = \frac{\partial u'_i}{\partial t} + \bar{u}_j \frac{\partial u'_i}{\partial x_j} + u'_j \frac{\partial u'_i}{\partial x_j} + u'_j \frac{\partial \bar{u}_i}{\partial x_j} + \frac{1}{\rho} \frac{\partial p'}{\partial x_i} - \frac{1}{\rho} \frac{\partial \sigma'_{ij}}{\partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} = 0 \quad (6)$$

and take the second moment (Speziale, 1991; Wilcox, 1993)

$$\overline{u'_i \mathcal{L}u'_j} + \overline{u'_j \mathcal{L}u'_i} = 0 \quad (7)$$

we obtain the transport equation

$$\frac{\partial \tau_{ij}}{\partial t} + \bar{u}_k \frac{\partial \tau_{ij}}{\partial x_k} = \bar{P}_{ij} + \Pi_{ij} + \frac{\partial \bar{D}'_{ijk}}{\partial x_k} - \varepsilon_{ij} + \nu \frac{\partial^2 \tau_{ij}}{\partial x_k \partial x_k} \quad (8)$$

where the right-hand side represents the rate of change of τ_{ij} produced by the turbulent production $\bar{P}_{ij}$, the pressure-strain rate correlation Π_{ij} , the turbulent diffusion $\bar{D}'_{ijk}$, the turbulent dissipation rate ε_{ij} , and the molecular diffusion. These terms are given by

$$\bar{P}_{ij} = -\tau_{ik} \frac{\partial \bar{u}_j}{\partial x_k} - \tau_{jk} \frac{\partial \bar{u}_i}{\partial x_k} \quad (8a)$$

$$\Pi_{ij} = \frac{p'}{\rho} \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right) \quad (8b)$$

$$\bar{D}'_{ijk} = - \left[\overline{u'_i u'_j u'_k} + \frac{p'}{\rho} \left(u'_i \delta_{jk} + u'_j \delta_{ik} \right) \right] \quad (8c)$$

$$\varepsilon_{ij} = 2\nu \overline{\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k}} \quad (8d)$$

These equations represent six independent tensor components that constitute an incompressible Reynolds stress transport equation. Unfortunately, these equations are not closed and require modelling of the third-order correlations, the pressure-velocity and pressure-strain rate correlations, and the turbulent dissipation rate. Because in the modelling of the turbulent diffusion term $\bar{D}'_{ijk}$ in incompressible flows the pressure-velocity correlation is commonly dropped and only the turbulent transport (triple-moment) term is modelled, the above transport equation requires three terms to be modelled: the pressure-strain correlation Π_{ij} , the turbulent diffusion $\bar{D}'_{ijk}$, and the dissipation rate ε_{ij} .

For the second-moment closures, a transport equation for the dissipation rate has been almost exclusively coupled with (8), although Wilcox (1988b) has proposed a multiscale model that couples the Reynolds stress transport equation to the specific dissipation rate ω (*turbulent dissipation rate/turbulent kinetic energy*). Nevertheless, the focus here will be exclusively on the

dissipation rate closures. To best interpret the closure modelling for the turbulent stress transport equations, partition the tensor dissipation rate into its isotropic and deviatoric parts

$$\varepsilon_{ij} = \frac{2}{3} \varepsilon \delta_{ij} + \varepsilon_{ij}^d \quad (9)$$

All attempts at modelling the deviatoric part of (9) to date have focused on a coupling with the Reynolds stress anisotropy (e.g. Hanjalić and Launder, 1976; Hallbäck *et al.*, 1990). In general, however, the deviatoric dissipation has been assimilated into the pressure-strain correlation Π_{ij} with the composite term

$$\Pi_{ij} = \Pi_{ij} - \varepsilon_{ij}^d \quad (10)$$

to be modelled. The isotropic dissipation rate is then explicitly retained in the τ_{ij} transport equation (8) and must be modelled. In both the Reynolds stress equation formulation and the two-equation formulation to follow, a transport equation for the isotropic dissipation rate is used. This transport equation for ε can be derived from the fluctuating momentum equation by taking the moment (Speziale, 1991)

$$\nu \frac{\partial u'_i}{\partial x_j} \frac{\partial (\mathcal{L} u'_i)}{\partial x_j} = 0 \quad (11)$$

A complex equation is obtained in which higher-order moments appear that all have to be modelled. The form of these individual terms can be found in Speziale (1991) or Wilcox (1993) (cf. Launder *et al.*, 1975) and can be grouped into production, destruction, and diffusion of dissipation contributions to the transport equation balance. The lack of rigour associated with the modelling of the terms in the dissipation rate equation is an obvious deficiency in the overall modelling of turbulent flows. Nevertheless, even with the associated problems, the form of the isotropic dissipation rate equation that is used is relatively standard. In the present context the high-Reynolds-number form of the modelled isotropic dissipation rate equation is simply written as

$$\frac{\partial \varepsilon}{\partial t} + \bar{u}_k \frac{\partial \varepsilon}{\partial x_k} = \bar{\mathcal{P}}_\varepsilon - \bar{\mathcal{D}}_\varepsilon + \bar{\mathcal{D}}'_\varepsilon + \nu \frac{\partial^2 \varepsilon}{\partial x_k \partial x_k} \quad (12)$$

with

$$\bar{\mathcal{P}}_\varepsilon = -C_{\varepsilon 1} \frac{\varepsilon}{K} \tau_{kl} \frac{\partial \bar{u}_k}{\partial x_l} \quad (12a)$$

$$\bar{\mathcal{D}}_\varepsilon = C_{\varepsilon 2} \frac{\varepsilon^2}{K} \quad (12b)$$

$$\bar{\mathcal{D}}'_\varepsilon = C_\varepsilon \frac{\partial}{\partial x_k} \left(\frac{K}{\varepsilon} \tau_{kl} \frac{\partial \varepsilon}{\partial x_l} \right) \quad (12c)$$

where the right-hand-side represents the production, destruction, turbulent diffusion, and viscous diffusion, respectively, and the coefficients $C_{\varepsilon 1}$, $C_{\varepsilon 2}$, and C_ε are closure constants. The values for $C_{\varepsilon 1}$ and $C_{\varepsilon 2}$ are set in conjunction with the pressure-strain closure, although the value of $C_{\varepsilon 2}$ is chosen close to 1.90 to match the decay rate of isotropic turbulence. Depending on the stress closure model and the values of the coefficients $C_{\varepsilon 1}$ and $C_{\varepsilon 2}$, the closure coefficient C_ε is generally set between the values 0.15 and 0.18 to have the correct log-law slope in the equilibrium boundary layer (see Launder *et al.*, 1975; Abid and Speziale, 1993).

Pressure-strain rate correlation models

One of the most active areas of research in Reynolds stress modelling is focused on the development of pressure-strain correlation models. This term is significant because it is of the same order as the production term, and it acts as a redistribution term between the Reynolds stress components. It acts on the anisotropy of the stress field in such a way as to diminish the difference between the normal stress components. While the number of such closures is rather plentiful, recent research has focused on a few such models that characterize those in use. This characterization has focused on the classification of the various models by their dependency on the Reynolds stress anisotropy tensor defined as

$$b_{ij} = \frac{\tau_{ij}}{2K} - \frac{1}{3}\delta_{ij} \quad (13)$$

In all of the commonly used second-moment closure models, Π_{ij} is modelled in the general form (e.g. Lumley, 1978; Reynolds, 1987) as

$$\Pi_{ij} = \varepsilon \mathcal{A}_{ij}(\mathbf{b}) + K \mathcal{M}_{ijkl}(\mathbf{b}) \frac{\partial \bar{u}_k}{\partial x_l} \quad (14)$$

where $\mathcal{A}_{ij}(\mathbf{b})$ and $\mathcal{M}_{ijkl}(\mathbf{b})$ are tensor functions of the anisotropy tensor and are related to integrals over the flow volume derived from the pressure Poisson equation. As has been recently pointed out by Speziale (1992b), the interpretation of each of these terms is exact in homogeneous turbulent flows but ambiguous in inhomogeneous turbulent flows. The $\mathcal{A}_{ij}(\mathbf{b})$ term is usually associated with the 'slow' relaxation of the turbulence toward isotropy, and the $\mathcal{M}_{ijkl}(\mathbf{b})$ term is usually associated with the 'rapid' response of the turbulence to imposed mean velocity gradients. This partitioning has its origins in the splitting of the turbulent pressure field p' into slow $p'^{(S)}$ and rapid $p'^{(R)}$ parts. The $p'^{(S)}$ part is the solution of a Poisson equation that involves gradients of the turbulent velocity field, and the $p'^{(R)}$ part is the solution of a Poisson equation involving the mean velocity gradients. The terminology originates from the observation that the slow part will only adjust as the turbulence itself adjusts, but the rapid part will adjust instantly through the mean velocity gradient. The reader is

encouraged to review the references just cited for details of the theoretical foundation of such models. Representative models that are linear, quadratic, and cubic in the anisotropy tensor will be presented in order to acquaint the reader with their form.

In the past 25 years, Launder and colleagues at Imperial College and UMIST have extended the formulation of Naot *et al.* (1970) to develop a variety of pressure-strain correlation models. The linear Launder, Reece and Rodi (1975) LRR model in its high-Reynolds-number form has been quite popular. In terms of the anisotropy tensor b_{ij} , it can be written as:

Launder, Reece and Rodi model

$$\begin{aligned} \Pi_{ij} = & -C_1 \varepsilon b_{ij} + C_2 K \bar{S}_{ij} + C_3 K (b_{ik} \bar{S}_{jk} + b_{jk} \bar{S}_{ik} - \frac{2}{3} b_{mn} \bar{S}_{mn} \delta_{ij}) \\ & + C_4 K (b_{ik} \bar{W}_{jk} + b_{jk} \bar{W}_{ik}) \end{aligned} \quad (15)$$

where

$$C_1 = 3.0, \quad C_2 = 0.8, \quad C_3 = 1.75, \quad C_4 = 1.31 \quad (16)$$

with

$$C_{\varepsilon 1} = 1.44, \quad C_{\varepsilon 2} = 1.90 \quad (16a)$$

and $\bar{W}_{ij} = \frac{1}{2}(\partial \bar{u}_i / \partial x_j - \partial \bar{u}_j / \partial x_i)$ is the rotation rate tensor (i.e. the antisymmetric part of the mean velocity gradient tensor). Note that all of the models to be discussed can be generalized to non-inertial frames by adding $\epsilon_{mji} \Omega_m$ (where Ω_m is the rotation rate of the non inertial frame relative to the inertial frame) to $\bar{W}_{ij}$. In addition to the LRR model, the isotropization-of-production (IP) model (Launder *et al.*, 1975; Gibson and Launder, 1978) also fits into this form with the following coefficients:

Isotropization-of-production model

$$C_1 = 3.6, \quad C_2 = 0.8, \quad C_3 = 1.2, \quad C_4 = 1.2 \quad (17)$$

with

$$C_{\varepsilon 1} = 1.44, \quad C_{\varepsilon 2} = 1.90 \quad (17a)$$

A point should be made about the closure constants C_1 – C_4 . Of the four coefficients, only two are independent. These added constants appear due to the writing of the pressure-strain model in terms of b_{ij} , $\bar{S}_{ij}$, and $\bar{W}_{ij}$ rather than the τ_{ij} tensor and the mean velocity gradient tensor, as was the case when the models were originally derived. However, the present notation is preferred because it shows the interrelationship between the various pressure-strain models. The values of the coefficient of the slow term C_1 were optimized by

comparisons with homogeneous shear flow (LRR model) and further modified for the IP model to better match free-shear flow data. In both the LRR and IP models, the coefficients (recall that $C_2 - C_4$ are not independent) are calibrated from homogeneous shear flow data, and rapid distortion of homogeneously strained initially isotropic turbulence data. As indicated earlier, the values for the closure coefficients in the dissipation rate equation (12) are optimized in conjunction with the respective pressure-strain-rate models. The value of $C_{\varepsilon 1}$ was originally determined (Hanjalić and Launder, 1972) from the properties of a constant stress layer adjacent to a wall at which a production equal dissipation balance occurs, but was later refined (Launder *et al.*, 1975) by computer optimization. The coefficient $C_{\varepsilon 2}$ was determined from the decay of grid-generated turbulence. Note that in comparisons with large-eddy simulations of homogeneous shear flow, as well as with experimental results, the IP model gives slightly better predictions of both the temporal evolution of the turbulent kinetic energy and the equilibrium values of the components of the stress anisotropy b_{ij} than the more complete model.

It should be noted that both the LRR and IP models which have been presented do not include the 'wall reflection' term which is necessary to insure the correct log-law behaviour in wall-bounded flows. It depends on distance from the wall, and as such does not fit into the classification structure presently being outlined. It needs to be included in wall-bounded calculations using the LRR and IP models and its effect is felt over a wide range of the boundary layer. The other models to be presented do not have this restriction which can be troublesome in actual calculations.

At the next level of complexity, the pressure-strain correlation models are *quadratic* in the stress anisotropy b_{ij} . These closures also have coefficients $C_1 - C_4$ that are functions of the invariants of b_{ij} , and depend as well on $\bar{P}/\varepsilon$ ($\bar{P} \equiv \bar{P}/2$). One such model is the Speziale, Sarkar and Gatski (1991) SSG model. In the SSG model, the quadratic behaviour was originally included as a higher-order correction to the slow part of the pressure-strain correlation, and the rapid part was quasilinear in the stress anisotropy. As such, the composite pressure-strain Π_{ij} model is:

Speziale, Sarkar and Gatski model

$$\begin{aligned} \Pi_{ij} = & -C_1 \varepsilon b_{ij} + C'_1 \varepsilon (b_{ik} b_{kj} - \frac{1}{3} b_{mn} b_{nm} \delta_{ij}) + C_2 K \bar{S}_{ij} \\ & + C_3 K (b_{ik} \bar{S}_{jk} + b_{jk} \bar{S}_{ik} - \frac{2}{3} b_{mn} \bar{S}_{mn} \delta_{ij}) + C_4 K (b_{ik} \bar{W}_{jk} + b_{jk} \bar{W}_{ik}) \end{aligned} \quad (18)$$

where

$$C_1 = 3.4 + 1.8 \frac{\bar{P}}{\varepsilon}, \quad C'_1 = 4.2, \quad C_2 = 0.8 - 1.30 (b_{ij} b_{ij})^{\frac{1}{2}}, \quad C_3 = 1.25, \quad C_4 = 0.40 \quad (19)$$

with

$$C_{\varepsilon 1} = 1.44, \quad C_{\varepsilon 2} = 1.83 \quad (19a)$$

These values are obtained by the calibration with homogeneous shear flows (with and without rotation), the decay of isotropic turbulence and return to isotropy of initially anisotropic turbulence, and the rapid distortion of homogeneously strained turbulence that is initially isotropic. The dependence on $\bar{P}/\epsilon$ is introduced into the coefficient C_1 in order to achieve an asymptotically consistent truncation of the Taylor series expansion for Π_{ij} . Consistent with the behaviour of the anisotropy tensor b_{ij} , this ratio also attains an equilibrium value in flows such as plane homogeneous shear (≈ 1.5) and the log layer of a channel flow ($= 1$).

The final level of pressure-strain closures to be discussed are models that are *cubic* in the stress anisotropy tensor. A popular representative is the Fu, Launder and Tselepidakis (1987) FLT model. In the previous section, the deviatoric part of the tensor dissipation was assimilated into the pressure-strain correlation, and the composite was modelled. This procedure was done for the two models just discussed. In the FLT model, however, the deviatoric part of the dissipation rate $D\varepsilon_{ij}$ is explicitly represented in the formulation and a model for Π_{ij} rather than Π_{ij} is formed.

Fu, Launder and Tselepidakis model

$$\begin{aligned} \Pi_{ij} = & -C_1 \varepsilon b_{ij} + C'_1 \varepsilon (b_{ik} b_{kj} - \frac{1}{3} b_{kl} b_{kl} \delta_{ij}) \\ & + C_2 K \bar{S}_{ij} + 1.2K (b_{ik} \bar{S}_{jk} + b_{jk} \bar{S}_{ik} - \frac{2}{3} b_{kl} \bar{S}_{kl} \delta_{ij}) \\ & + \frac{26}{15} K (b_{ik} \bar{W}_{jk} + b_{jk} \bar{W}_{ik}) - \frac{112}{5} K [II (b_{ik} \bar{W}_{jk} + b_{jk} \bar{W}_{ik})] \\ & + \frac{4}{5} K (b_{ik} b_{kl} \bar{S}_{jl} + b_{jk} b_{kl} \bar{S}_{il} - 2b_{ik} \bar{S}_{kl} b_{lj} - 3b_{kl} \bar{S}_{kl} b_{ij}) \\ & + \frac{4}{5} K (b_{ik} b_{kl} \bar{W}_{jl} + b_{jk} b_{kl} \bar{W}_{il}) - \frac{168}{5} K (b_{ik} b_{kl} \bar{W}_{lm} b_{mj} + b_{jk} b_{kl} \bar{W}_{lm} b_{mi}) \end{aligned} \quad (20)$$

and

$$\varepsilon_{ij} = \frac{2}{3} \varepsilon \delta_{ij} + D \varepsilon_{ij} = \frac{2}{3} \varepsilon \delta_{ij} + 2(1 - \sqrt{F}) b_{ij} \varepsilon \quad (21)$$

where

$$F = 1 + 9II + 27III \quad (22)$$

$$II = -\frac{1}{2}b_{ij}b_{ij}, \quad III = \frac{1}{3}b_{ij}b_{jk}b_{ki} \quad (23)$$

$$C_1 = -120II\sqrt{F} - 2\sqrt{F} + 2, \quad C'_1 = 144II\sqrt{F}, \quad C_2 = 0.8 \quad (24)$$

with

$$C_{\varepsilon 1} = 1.45, \quad C_{\varepsilon 2} = 1.90 \quad (25)$$

While this model is tensorially more complex than either of the other two, its calibration is similar and is based on comparisons with the same benchmark flows used in the previous two calibrations. In addition, it also satisfies the constraint of realizability (Schumann, 1977; Lumley, 1978), which requires that a Reynolds stress model yield non-negative component energies. Because insufficient space is available here to discuss the details of the formulation, the interested reader is referred to the paper by Fu *et al.* (1987) for more details of this model, and to the paper of Speziale *et al.* (1994) for a discussion of the realizability aspects of the model.

Recently, Ristorcelli *et al.* (1994) developed a 2D MFI model that is similar in structure to the models just described, keeping terms cubic in the anisotropy tensor. However, two notable differences are associated with this model: (i) the coefficients are functions of the state of the flow, with the functional form deduced from mathematical constraints satisfied by the rapid pressure term and (ii) the model is also frame indifferent in the two-dimensional limit. This latter constraint insures that the model is consistent with the Taylor–Proudman theorem (Speziale, 1989); as such, the model is expected to perform well in situations where the flow is two dimensionalized by various body forces, such as buoyancy or rotation, or in situations where streamline curvature is important. Whether or not building more physics and mathematics into the model will actually produce a representation capable of successfully computing these more complex flows remains to be seen.

Some coordinated attempts have been made to assess the performance of the various pressure–strain models. One approach has been to simply solve the full Reynolds stress closure equations for a specific flow and compare the predictive capability of the models tested. The other approach has been to take either experimental or direct simulation data and calculate the pressure–strain correlation from the data and compare it with the various model representations using the same database.

In the former approach, practical engineering flows that are sufficiently simple to compute, and in which the results are not tainted by either numerical or model form ambiguities when applied, are required. Flows which fit these criteria are two-dimensional free shear flows such as jets, wakes and mixing layers, as well as two-dimensional flat-plate boundary-layer flows. More complex flows, such as separated wall-bounded flows, flows with curvature, and flows with imposed pressure gradients, are a few examples where numerical or

model form ambiguities can produce misleading results. Nevertheless, organized attempts at such tests are over 25 years old and date back to the Stanford 1968 competition on computation of two-dimensional boundary layers (Kline *et al.*, 1969); these earlier attempts were followed by the Stanford 1980/1981 conference on complex turbulent flows (Kline *et al.*, 1982). A similar endeavour on collaborative testing of turbulent models (Bradshaw *et al.*, 1994) was initiated in 1991.

Simpler, more generic flows, such as homogeneous shear both with and without rotation, can be calculated in an unambiguous way because the turbulent transport is zero and the mean velocity gradient is constant. Such numerically generated flows, where the turbulence is statistically homogeneous in the streamwise and transverse directions, are not simply lagrangian descriptions of a corresponding laboratory experiment. Experiments in uniformly sheared turbulence have been performed for some time; one of the most recent is that of Tavoularis and Karnik (1989). This and previous studies have shown that the uniformly sheared turbulence can be a close, although not exact, representation of the homogeneous turbulence generated by the numerical experiments, and can display the same energetic and dissipation growth rates.

These homogeneous shear flows in equilibrium have a common theoretical connection to the log layer in channel flow (Abid and Speziale, 1993) in that a second-order closure model will yield the same equilibrium values for the anisotropy b_{ij} in both flows. As such, they are important indicators of the performance level of the models when applied to practical wall-bounded flows. Direct numerical simulations and large-eddy simulations of the homogeneous shear flows have been performed to compare the predictive performance of the pressure-strain models directly. Recently, Speziale *et al.*, (1990, 1991) have tested the LRR, SSG, and FLT models in homogeneous shear flows with and without rotation. The results showed that the SSG model outperforms the linear LRR and cubic FLT models in predicting the evolution of turbulent kinetic energy; however, none of the models was able to predict accurately the evolution of turbulent kinetic energy for the most energetic (unstable) homogeneous shear flow case with rotation.

In the latter approach, in which experimental and/or DNS data are used, the results are less ambiguous but require extensive data to compute the various pressure-strain correlations and the model representation. Schwarz and Bradshaw (1994) have used experimental data from a three-dimensional boundary layer flow to compute the behaviour of a variety of pressure-strain models. They found that the quadratic SSG model predictions were generally closest to the data but differed little from the linear LRR model. The cubic FLT model predicted the normal stresses well, but did not predict the τ_{12} shear stress as well as the LRR model. Some of their results are shown in Figures 1 and 2 for the streamwise normal stress component and the τ_{12} shear stress component. In these figures, the behaviour of the Π_{11} and Π_{12} correlations are compared with

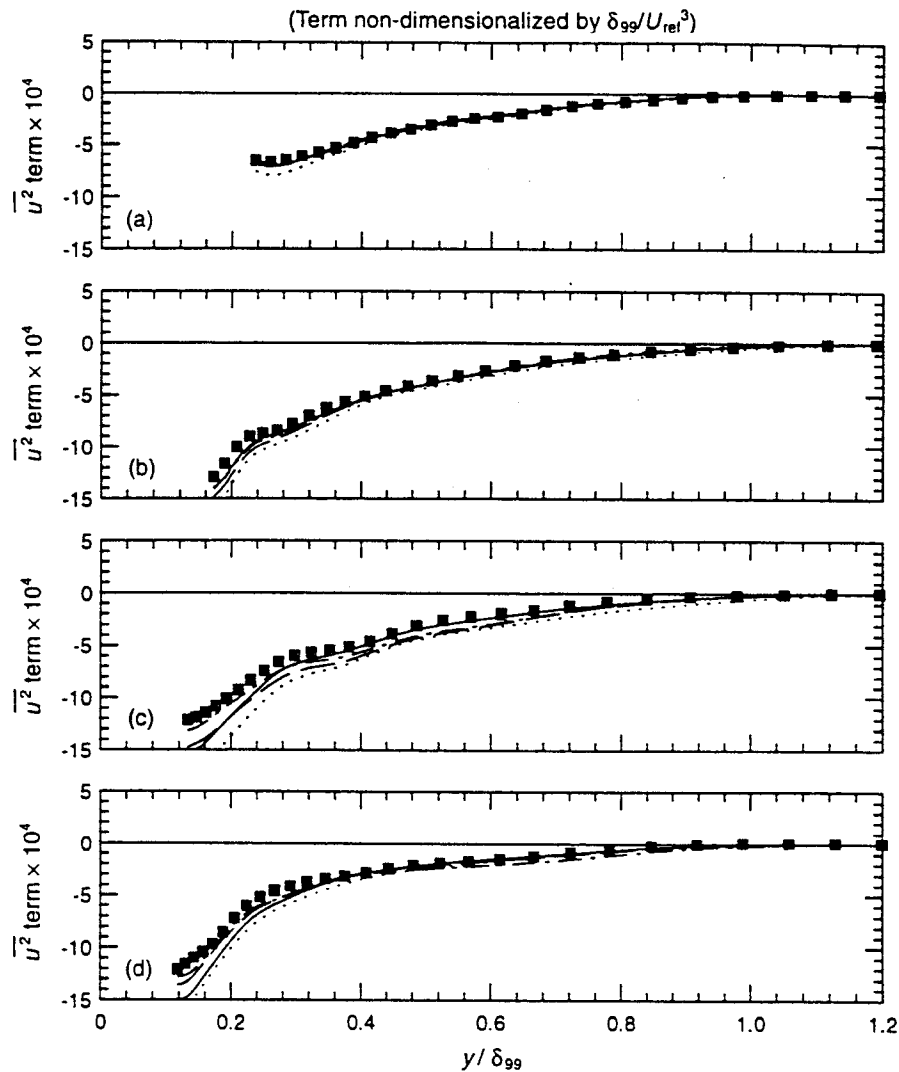


Figure 1 Comparisons of models for the Π_{11} pressure-strain term (Schwarz and Bradshaw, 1994). (a) Station 1: $x = 978$ mm. (b) Station 7: $x = 1775$ mm. (c) Station 13: $x = 2415$ mm. (d) Station 18: $x = 2948$ mm. Key: ■, data; ···, LRR; ---, SSCG; —, SL/CL; ---, FLT.

experimental results at various downstream locations in the three-dimensional flow. Also included in the comparison is the non-linear Shih and Lumley (1985) SL rapid model, coupled with the Choi and Lumley (1984) slow model. The model is not given here because the details of the formulation are complex; however, the authors attempt to utilize a strong form of the realizability

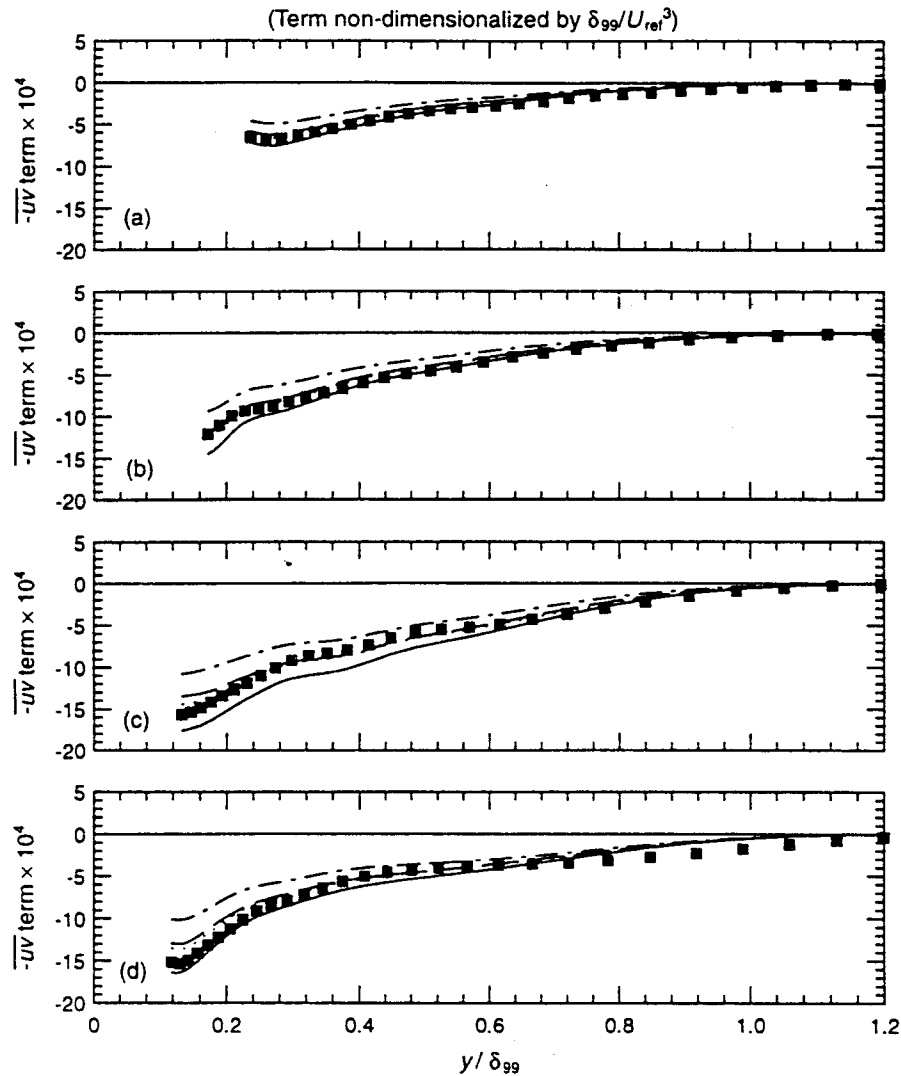


Figure 2 Comparisons of models for the Π_{12} pressure-strain term (Schwarz and Bradshaw, 1994). (a) Station 1: $x = 978$ mm. (b) Station 7: $x = 1775$ mm. (c) Station 13: $x = 2415$ mm. (d) Station 18: $x = 2948$ mm. Key: ■, data; ···, LRR; ---, SSG; —, SL/CL; -·-, FLT.

constraint. This constraint requires that when a principal Reynolds stress component vanishes, its time rate of change must vanish and its second derivative must be positive. Recently, this model has come under closer scrutiny by Speziale *et al.* (1994), who found that the model is not realizable and who questioned the use of the realizability constraint on closure models based on

near-equilibrium conditions. Speziale *et al.* (1992b) have performed a comparative study using DNS data of homogeneous turbulence to assess the performance of the pressure-strain models. The models tested did not include the linear LRR model; however, the quadratic SSG model and the cubic FLT model were tested. They concluded that both the SSG and FLT models performed equally well in predicting the various pressure-strain correlations. Another recent study by Younis *et al.* (1994) has tested the IP and SSG models in three complex turbulent flows: rotating channel, boundary layer over convex curved surface, and boundary layer over an infinite swept wing. In these flows the SSG model performed as well as or better than the IP model.

The correlation models presented are not meant to be all inclusive; however, they are put forward as typical of the functional form encountered for present day pressure-strain models.

Turbulent diffusion models

A model for the turbulent diffusion term $\bar{D}_{ijk}'$ in (8c), with the pressure-velocity correlation neglected, has traditionally been developed, although Lumley (1978) has made a theoretical estimate that the pressure-velocity correlation is about 20% percent of its triple-velocity correlation counterpart. Nevertheless, this term has received very little scrutiny in numerical simulations and experiments and has not been explicitly modelled in incompressible flows. The gradient transport hypothesis assumes that turbulent diffusion behaves analogously to molecular diffusion and, thus, a distinct scale separation exists between the two diffusion processes.

The full second-moment closures use the gradient transport hypothesis to model the turbulent diffusion term $\bar{D}_{ijk}'$. The forms are more complex, however, because the individual stress components are utilized in the resulting models. The three most common models used in conjunction with the second-moment closures fall into the following general functional form (Speziale, 1991)

$$\bar{D}_{ijk}' = d_{ijklmn} \frac{\partial \tau_{lm}}{\partial x_n} \quad (26)$$

where d_{ijklmn} can be a function of τ_{ij} .

Daly and Harlow (1970) (DH) proposed the rather simple model

$$\bar{D}_{ijk}' = C_s \frac{K}{\varepsilon} \tau_{kl} \frac{\partial \tau_{ij}}{\partial x_l} \quad (27)$$

where $C_s = 0.22$ is a diffusion constant determined from model optimization. While (27) is straightforward, it suffers from the obvious deficiency that it is not invariant under the permutation of indices. A second, less serious deficiency is the fact that in two-dimensional thin shear flows where the dominant diffusion is in the cross stream direction, the model gives an isotropic effective diffusion coefficient; although in more complex flows the model is anisotropic. It is important to recognize that these closure constants for the turbulent diffusion

models are determined in the calibration process of the entire second-moment model. For example, the value 0.22 is the value recommended by Daly and Harlow (1970), but Launder *et al.* (1975) in the LRR model found an optimized value of 0.25 for C_s in their calibration studies. Therefore, it is necessary to use the value of C_s associated with each model for $\bar{D}'_{ijk}$ as an initial estimate; for each second-moment closure model (fully modelled stress transport equation plus dissipation rate model), a unique set of closure coefficients is determined. An example of this modelling *in toto* is the LRR model, where the modelled terms in the stress transport equation and the dissipation rate equation have been optimized together.

A much more complicated expression has been proposed by Hanjalić and Launder (1972) (HL), who modelled the transport equation for the third-order moments and then extracted an algebraic expression from the modelled equation by neglecting the transport terms and assuming a gaussian relation to link the fourth-order correlations to the second-order correlations. Their invariant expression for $\bar{D}'_{ijk}$ is

$$\bar{D}'_{ijk} = C_s \frac{K}{\varepsilon} \left(\tau_{il} \frac{\partial \tau_{jk}}{\partial x_l} + \tau_{jl} \frac{\partial \tau_{ki}}{\partial x_l} + \tau_{kl} \frac{\partial \tau_{ij}}{\partial x_l} \right) \quad (28)$$

The value for C_s was originally chosen by Hanjalić and Launder (1972) (HL) as 0.08; however, later LRR chose a value of 0.11. In both cases, the value was chosen to optimize the predictive capability of the *entire* closure model.

The final model to be presented is the Mellor and Herring (1973) (MH) model. This model is an isotropized version of the HL model and is given by

$$\bar{D}'_{ijk} = C_s \frac{K^2}{\varepsilon} \left(\frac{\partial \tau_{jk}}{\partial x_i} + \frac{\partial \tau_{ki}}{\partial x_j} + \frac{\partial \tau_{ij}}{\partial x_k} \right) \quad (29)$$

where the model constant C_s is taken as two-thirds of the value chosen for the HL model. This model also retains the proper coordinate invariance properties.

Other more complicated models have also been proposed. Lumley (1978) proposed a model that contains the HL model as a tensorial basis. He also proposed an explicit model for the pressure-velocity diffusion to be used in conjunction with the triple-moment model. Recently, Magnaudet (1993) has proposed a model that is consistent with the two-dimensional turbulence encountered near a wall and at a free surface. This model is also quite complex and has not been tested for a variety of engineering flows. The interested reader is also referred to the recent review of Hanjalić (1994) for further commentary.

Recently, Demuren and Sarkar (1993) tested the performance of the DH, HL, and MH models in plane channel flow and found that the MH model gives the best performance because of its ability to properly model the relaxation to isotropy in the outer part of the layer (Figure 3). Schwarz and Bradshaw (1994) have compared the DH and HL models, as well as the Lumley model, in a three-dimensional boundary layer flow. They concluded that all three models

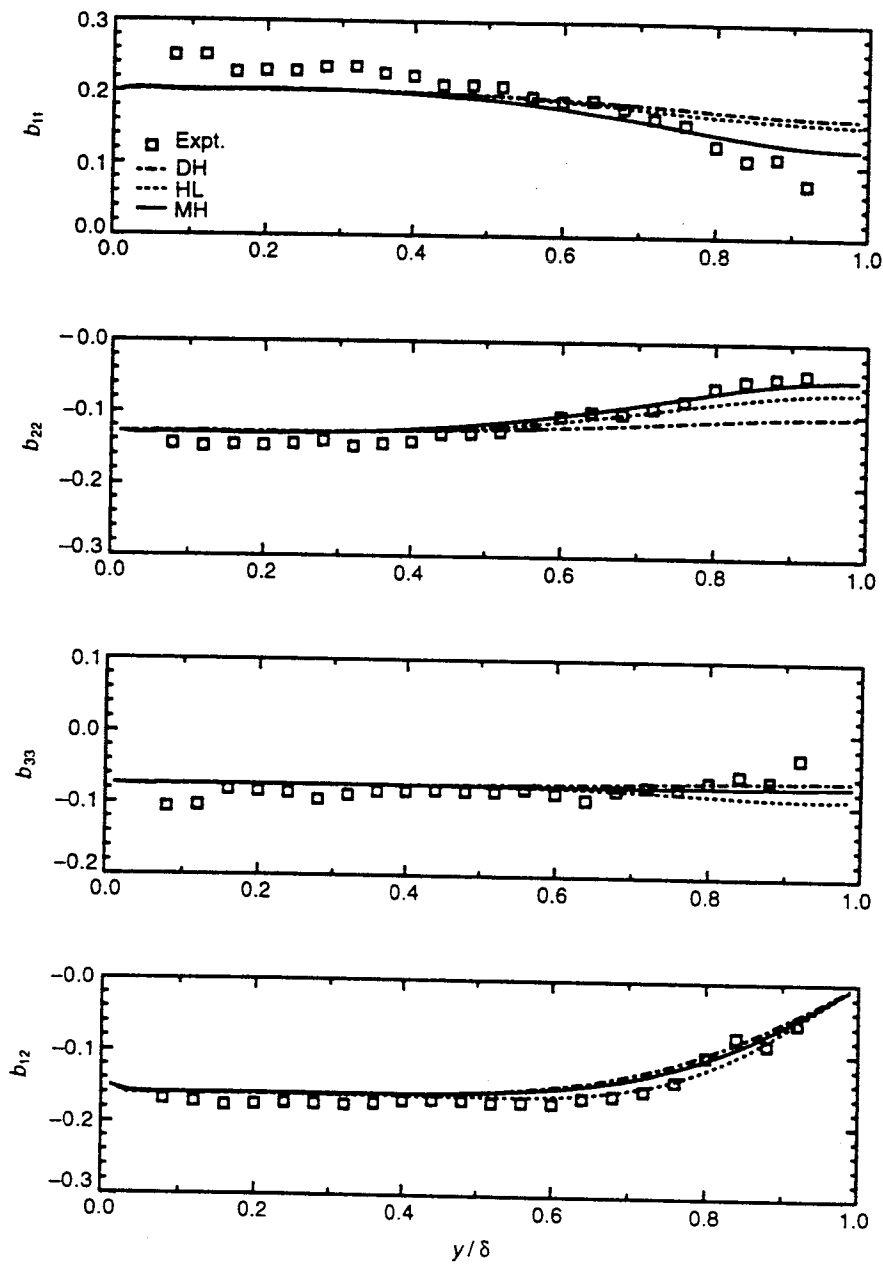


Figure 3 Comparison of components of the Reynolds stress anisotropy tensor, computed with SSG pressure-strain model and three diffusion models, with experimental data of Laufer (1951).

performed reasonably well; the Lumley model outperformed the other models in some component predictions.

2 Two-equation models

As the previous subsection has shown, the Reynolds stress formulation is capable of yielding the individual components of the second-moment or Reynolds stress tensor at the expense of rather complex differential equations and the need to model higher-order correlations. This complexity is eased somewhat by only looking at the trace of the τ_{ij} equations. To continue with the terminology introduced in Section II.B.1, the trace of τ_{ij} is designated by $\tau_{ii} (\equiv 2K)$, and the turbulent kinetic energy is given by ρK . For brevity and consistency, because the density ρ is constant, the quantity K will sometimes be termed the turbulent kinetic energy. This equation is constructed by taking the contraction of (8). Because the pressure-strain redistribution term is traceless ($\Pi_{ii} = 0$), the equation is less difficult to close because only diffusion models must be found. The disadvantage of this formulation is that the individual stress components cannot be determined directly from the solved equation set. The turbulent kinetic energy equation is

$$\frac{\partial K}{\partial t} + \bar{u}_j \frac{\partial K}{\partial x_j} = \bar{P} + \frac{\partial \bar{D}_j'}{\partial x_j} - \epsilon + \nu \frac{\partial^2 K}{\partial x_j \partial x_j} \quad (30)$$

where the right-hand-side represents the transport of K by the turbulent production $\bar{P} \equiv \bar{P}_{ii}/2$, the turbulent diffusion $\bar{D}_j' \equiv \bar{D}_{ij}'/2$, the isotropic turbulent dissipation rate, $\epsilon \equiv \epsilon_{ii}/2$, and the viscous diffusion. Consistent with the definitions used in the Reynolds stress formulation, the production, diffusion, and dissipation terms are given by

$$\bar{P} = -\tau_{ij} \frac{\partial \bar{u}_i}{\partial x_j} \quad (30a)$$

$$\bar{D}_j' = - \left[\frac{1}{2} \overline{u_i' u_i' u_j'} + \frac{\overline{p' u_i'}}{\rho} \delta_{ij} \right] \quad (30b)$$

$$\epsilon = \nu \overline{\frac{\partial u_i'}{\partial x_j} \frac{\partial u_i'}{\partial x_j}} \quad (30c)$$

respectively. Even with this formulation, the individual stress components, which are required in the production term of the kinetic energy equation, still must be known. Depending on the formulation of the mean-flow problem, the individual stress components appear explicitly in the Reynolds-averaged Navier-Stokes equations. In the two-equation formulation, the explicit appearance of the individual stress components is generally not the case

because the effect of the turbulence on the mean flow is usually accounted for through an eddy viscosity.

Dimensional analysis considerations dictate that the eddy viscosity ν_t be given by the product of a turbulent velocity scale and a turbulent length scale. In a two-equation K - ϵ formulation, these are proportional to $K^{1/2}$ and $K^{3/2}/\epsilon$, respectively. Thus, the eddy viscosity ν_t is given by the relation

$$\nu_t = C_\mu \frac{K^2}{\epsilon} \quad (31)$$

where $C_\mu = 0.09$ is the usual value assumed.

Consistent with the simplification that results from the two-equation formulation, a simple form for the turbulent transport $\bar{D}_{ijk}'$ is usually used in the kinetic energy equation. Although the models required for the Reynolds stress transport equation are more complex, the underlying gradient transport hypothesis that is used is also applied to the simpler turbulent diffusion model for the kinetic energy turbulent transport term. The turbulent diffusion model is given by

$$\bar{D}_j' = \frac{\nu_t}{\sigma_K} \frac{\partial K}{\partial x_j} \quad (32)$$

where σ_K is a constant that acts like an effective Prandtl number for diffusion. In most cases, σ_K is assumed to be unity. With this model for the turbulent diffusivity, both the turbulent and molecular diffusion terms can be combined into a gradient transport model with effective viscosity $\nu + \nu_t/\sigma_K$. The resulting simple form of the modelled turbulent kinetic energy equation is the obvious appeal of the formulation.

Note that in both the kinetic energy and Reynolds stress equations, the turbulent dissipation rate is chosen as the fundamental scale variable, and, as was just shown, can be combined with the turbulent stresses to form both turbulent length and time scales. Other scale variables, which will be discussed shortly, can be derived from this quantity; however, for the most part, the dissipation rate is the scale variable used.

The modelled transport equation for the isotropic dissipation rate is once again given by (12), but now the turbulent diffusion term $\bar{D}_\epsilon'$ in the equation can be simplified relative to the form used in the Reynolds stress transport formulation. The analogue to the turbulent diffusion term that appears in (12) is the simpler

$$\bar{D}_\epsilon' = \frac{\partial}{\partial x_k} \left(\frac{\nu_t}{\sigma_\epsilon} \frac{\partial \epsilon}{\partial x_k} \right) \quad (33)$$

where σ_ϵ is a constant that acts like an effective Prandtl number for dissipation

diffusion. In most cases, σ_ϵ is taken as 1.3 in order to ensure the correct log-law behaviour in boundary layer flows.

Although the K - ϵ formulation is the most popular two-equation model, it would be remiss not to mention that several other two-equation models have been formulated. Most of these other formulations have used alternatives to the dissipation-rate-scale equation. Kolmogorov (1942) was the first to propose the dissipation rate per unit kinetic energy ω or specific dissipation rate as a scale variable. Saffman (1970) offered an improved alternative version, and Wilcox (see Wilcox (1993) for details and associated references) has refined and applied the model in the last 20 years to such a wide variety of incompressible and compressible flows that it is the most widely used alternative to the dissipation rate equation. Speziale *et al.* (1992a) have offered a K - τ model that has the correct asymptotic behaviour near the wall, and have also proposed the use of a transport equation for the time scale variable $\tau = K/\epsilon$, which also has the correct near-wall asymptotics and is not singular at the wall. Two other alternatives have been the K - Kl (Wolfshtein, 1970; Mellor and Herring, 1973), and the K - $K\tau$ model (Zeierman and Wolfshtein, 1986). Both of these models use the two-point velocity correlation tensor (in space and time, respectively) and the associated transport equations to extract the appropriate scale equations.

In the last few years, renormalization group (RNG or RG) methods have been employed (Yakhot and Orszag, 1986) to develop alternate two-equation K - ϵ models for turbulent flows. The approach has since been further developed, and a variety of validation calculations performed (e.g. Yakhot *et al.*, 1992). Although the details of the formulation are outside the scope of this chapter, note that the Yakhot and Orszag approach is formally based on an ϵ -RNG methodology, in which a small parameter ϵ is introduced into the exponent of a forcing correlation function, which is in turn introduced into the momentum equation. The theory takes into account the non-local (spectral) interactions that occur in the flow. Recently, an alternative *Recursion*-RNG approach has been used to develop a modification to the usual Boussinesq formulation used in conjunction with the two-equation K - ϵ formulation (Zhou *et al.*, 1994a, b). The formulation does not rely on an ϵ expansion and accounts for local (spectral) interactions that occur in the flow. The technique has not been extended to developing a corresponding *Recursion*-RNG two-equation model. Both approaches may provide advantages over the more traditional two-equation K - ϵ formulations, but more widespread use and validation are required to assess their general applicability.

Much more could be discussed about the two-equation models and, in particular, the K - ϵ model because of the ease with which they can be applied and the widespread use they have enjoyed. Space limitations prevent such a discussion, but the interested reader is referred to the recent book by Mohammadi and Pironneau (1994), which is devoted entirely to the K - ϵ turbulence model.

Boussinesq approximation

Normally, in all of the two-equation formulations that have been mentioned, τ_{ij} components are *approximated* by analogy with a newtonian type of (linear) constitutive relationship between the turbulent stress and the mean strain-rate tensor $\bar{S}_{ij}$. This Boussinesq type of eddy-viscosity approximation is given by

$$\tau_{ij} = \frac{2}{3}K\delta_{ij} - 2\nu_t\bar{S}_{ij} \quad (34)$$

Equation (34) can also be obtained formally from a simple continuum mechanics approach (Speziale, 1991) by assuming that the stress tensor τ_{ij} is a tensor function of the mean velocity gradient and by assuming single turbulent length and time scales that are spectrally distinct from the larger mean scales. This single-scale assumption is the underlying basis contained in all standard turbulent closure models that are discussed. Attempts have been made at multiscale closures (e.g. Hanjalić *et al.* (1979) and Wilcox (1986) with moderate success); however, their usage has been generally confined to the originators.

Although the two-equation formulation with a Boussinesq constitutive relation is appealing, its omission of any anisotropic eddy-viscosity effects can lead to serious predictive deficiencies. A simple square-duct flow is an ideal example. The two-equation formulation with an isotropic eddy viscosity predicts no secondary streamline pattern; however, as Figure 4 shows, the qualitatively correct pattern is clearly predicted by the Reynolds stress model because of the proper accounting of the normal stress differences in the flow.

Explicit algebraic stress models

The two-equation formulation is appealing because it provides the transport equations for the turbulence that help account for some nonlocal and history effects but, at the same time, is not overly computationally intensive. One drawback, however, is the Boussinesq constitutive relationship (34), which assumes an isotropic eddy viscosity that is unable to account for strong anisotropies in the turbulence and results in poor predictions in flows with secondary motions. In addition, body forces such as an imposed rotational field are also not properly accounted for with the Boussinesq relationship. Furthermore, as noted, the pressure-strain correlation term has disappeared in the two-equation formulation, which precludes the effects of intercomponent transfer between τ_{ij} components. An effective compromise between the full second-moment closure and the two-equation model is the algebraic stress model (ASM). This approach extracts an algebraic relationship between the turbulent Reynolds stress and mean velocity field from the Reynolds stress closures discussed in Section II.B.1.

Algebraic stress models are derivable from equilibrium hypotheses imposed on both convective and diffusive effects. The convective equilibrium hypothesis

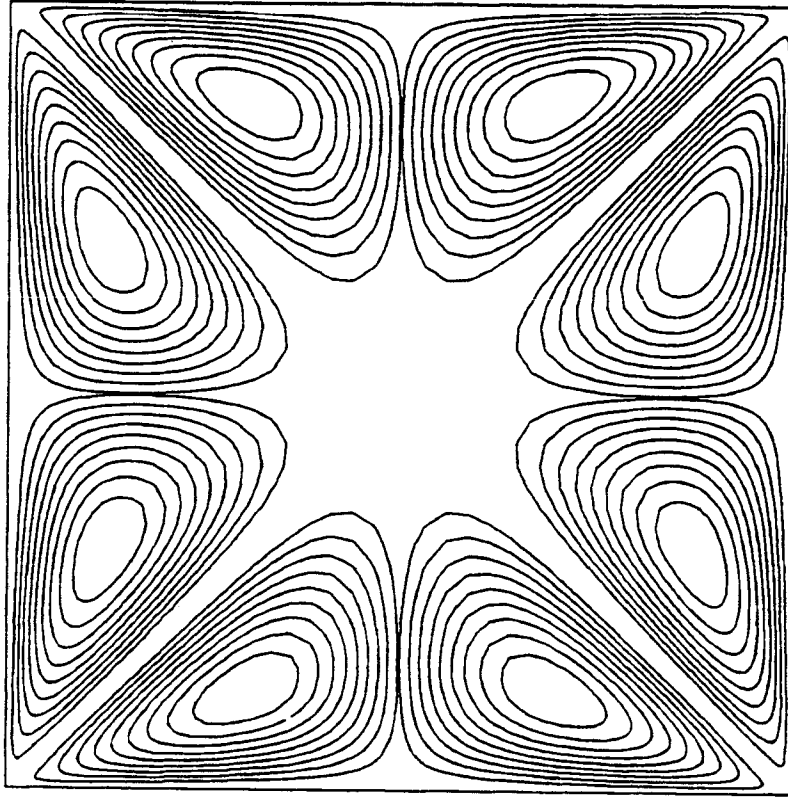


Figure 4 Secondary flow streamlines in a square duct with SSG Reynolds stress model (So, private communication).

is represented through the stress anisotropy tensor b_{ij} given by

$$\frac{Db_{ij}}{Dt} \equiv \frac{D\tau_{ij}}{Dt} - \frac{\tau_{ij}}{K} \frac{DK}{Dt} = 0 \quad (35)$$

and the equilibrium hypothesis for the diffusive terms is given by

$$\frac{\partial}{\partial x_k} \left(\bar{D}_{ijk}' + \nu \frac{\partial \tau_{ij}}{\partial x_k} \right) = 0 \quad (36)$$

If we use these constraints as well as the definition of the stress anisotropy given in (13), and consider the fact that the mean velocity gradient is the sum of the strain- and rotation-rate tensors, then (35) can be expanded as

$$\begin{aligned} (\bar{P} - \epsilon)b_{ij} = & -\frac{2}{3}K\bar{S}_{ij} - K(b_{ik}\bar{S}_{jk} + b_{jk}\bar{S}_{ik} - \frac{2}{3}b_{mn}\bar{S}_{mn}\delta_{ij}) \\ & -K(b_{ik}\bar{W}_{jk} + b_{jk}\bar{W}_{ik}) + \frac{1}{2}\Pi_{ij} \end{aligned} \quad (37)$$

This relation has been the traditional starting point for the algebraic stress models that have been developed.

Rodi (1976) was the first to implement an *implicit* algebraic relationship with the LRR model. It led to a *non-constant* coefficient C_μ in the eddy-viscosity definition (31), which was a function of the turbulent production and dissipation rate. Unfortunately, the potential of the formulation did not fully materialize because of the implicit nature of the algebraic relationship. At each iteration (time) step, an additional inner iteration is required to solve for the appropriate τ_{ij} component. This iterative process can cause solution divergence, and increased numerical overhead because of the extensive matrix inversions required at the different iteration levels. With the ASM formulated in this way, it is more appropriately viewed as a subset of the Reynolds stress formulation rather than a subset of the two-equation formulation.

Recently, interest has been renewed in algebraic stress models, as well as in the related area of non-linear two-equation models; the intent has been development of higher-order constitutive relations between the Reynolds stress tensor and the mean velocity gradient field. In this approach, the emphasis shifts from an alternative Reynolds stress formulation to an improved two-equation formulation. Gatski and Speziale (1993) have developed an *explicit* constitutive relationship for the Reynolds stress tensor with the ideas of Pope (1975), who proposed a tensorial polynomial expansion. The development is generalizable to a class of pressure-strain correlation models that are linear in the anisotropy tensor b_{ij} . This includes the linear LRR and IP models discussed previously, as well as linearized versions of such non-linear models as the SSG model. A 10-term polynomial representation, applicable to three-dimensional flows, which is quartic in products of the strain-rate and rotation-rate tensors, is found. For a two-dimensional flow, a general quadratic constitutive relationship can be written as

$$\tau_{ij} = \frac{2}{3}K\delta_{ij} - \nu_t^* \left(S_{ij}^* + \frac{K}{\varepsilon} [(S_{ik}^* W_{kj}^* + S_{jk}^* W_{ki}^*) - 2(S_{ik}^* S_{kj}^* - \frac{1}{3}S_{kl}^* S_{kl}^* \delta_{ij})] \right) \quad (38)$$

where

$$\nu_t^* = \frac{6(1 + \eta^2)\alpha_1}{3 + \eta^2 + 6\zeta^2\eta^2 + 6\zeta^2} \frac{K^2}{\varepsilon} \quad (39)$$

$$\eta = \frac{K}{\varepsilon} (S_{ij}^* S_{ij}^*)^{\frac{1}{2}}, \quad \zeta = \frac{K}{\varepsilon} (W_{ij}^* W_{ij}^*)^{\frac{1}{2}} \quad (39a)$$

$$S_{ij}^* = \frac{1}{2}g(2 - C_3)\bar{S}_{ij} \quad (39b)$$

$$W_{ij}^* = \frac{1}{2}g(2 - C_4)\bar{W}_{ij} \quad (39c)$$

with $g = (C_1/2 - 1 + \mathcal{P}/\epsilon)^{-1}$, and $\alpha_1 = (C_2 - \frac{4}{3})/(C_3 - 2)$. The coefficients are obtained from the general linear pressure-strain model (15).

Gatski and Speziale (1993) have compared the present formulation with other nonlinear two-equation models such as those by Yoshizawa (1984), Speziale (1987), and Rubinstein and Barton (1990). This comparison has shown that these models have a similar basis and should represent the same essential physical behaviour when implemented. Taulbee *et al.* (1994) has also derived algebraic stress models but has modified the coefficients proposed in the original LRR model to be able to develop a truncated series expansion that requires no regularization as the more general formulation just described. Unfortunately, this recalibration results in b_{33} anisotropy levels for homogeneous shear flow that are essentially zero, which contradicts both the experimental and numerical simulation results.

3 One-equation models

The next level in this hierarchy of turbulent closure models results in only the solution of a single transport equation. As outlined in Section I, the physical motivation behind the development of the closure schemes is to be able to describe the turbulence through both characteristic velocity and length scales. Both the Reynolds stress and two-equation models just described provide this information through combinations of the Reynolds stress (kinetic energy) field and the dissipation rate field. At the one-equation level of closure and below, insufficient information from the transport equation to be solved is available to determine both quantities.

The earliest one-equation models date back to Prandtl and involve the turbulent kinetic energy equation (velocity scale) and a prescription for the turbulent length scale. Modern-day approaches have evolved beyond this formulation to the solution of transport equations for the turbulent Reynolds number or the turbulent eddy viscosity (velocity scale $\times$ length scale). These more recent formulations will be discussed here, and the interested reader can refer to the text by Wilcox (1993) for a more complete historical record.

Baldwin and Barth (1991) devised a one-equation model that has its origins in the K - ϵ two-equation formulation. A turbulent eddy viscosity $\nu \bar{R}_t = K^2/\epsilon$ is defined, as well as its transport equation, which is derivable from the modelled K - ϵ equations (30) and (12) (with (33)). This equation is further modified by simplifying the form of the turbulent diffusion term and neglecting the two coupling terms. The result is a self-consistent one-equation model for $\nu \bar{R}_t$. The neglect of the coupling terms is formally valid in equilibrium flows where turbulent energy production equals dissipation. For the present introductory purposes, only a relatively simple form of the model will be presented; such factors as near-wall damping and intermittency effects on the form of the model

are excluded but can be found in the original paper. The resulting equation is

$$\frac{\partial(\nu \bar{R}_t)}{\partial t} + \bar{u}_k \frac{\partial(\nu \bar{R}_t)}{\partial x_k} = (C_{\varepsilon 2} - C_{\varepsilon 1}) \sqrt{(\nu \bar{R}_t) \mathcal{P}} + \left(\nu + \frac{\nu_t}{\sigma_\varepsilon} \right) \frac{\partial^2(\nu \bar{R}_t)}{\partial x_k \partial x_k} - \frac{1}{\sigma_\varepsilon} \frac{\partial \nu_t}{\partial x_k} \frac{\partial(\nu \bar{R}_t)}{\partial x_k} \quad (40)$$

where $\nu_t = C_\mu(\nu \bar{R}_t)$, and

$$C_{\varepsilon 1} = 1.2, \quad C_{\varepsilon 2} = 2.0, \quad C_\mu = 0.09, \quad \sigma_\varepsilon = \frac{(C_{\varepsilon 2} - C_{\varepsilon 1}) \sqrt{C_\mu}}{\kappa^2} \quad (41)$$

with the Von Karman constant $\kappa = 0.41$. Both $C_{\varepsilon 1}$ and $C_{\varepsilon 2}$ have values somewhat different from their values in the two-equation models discussed previously. The decay coefficient $C_{\varepsilon 2}$ is set to 2 to correspond to early experimental data on the decay of grid turbulence that showed a decay inversely proportional to downstream distance. Later experiments showed a more rapid decay that is the basis for the decay coefficient utilized in most of the two-equation models. With the decay coefficient fixed, the coefficient $C_{\varepsilon 1}$ is calibrated to match the skin-friction coefficient for incompressible flow over a flat plate.

In this one-equation formulation, the turbulence field is coupled to the mean field only through the turbulent eddy viscosity, which appears as part of an effective viscosity $(\nu + \nu_t)$ in the diffusion term of the Reynolds-averaged Navier-Stokes equation.

Other contemporary one-equation models using the eddy viscosity have been proposed. One is the Gulyaev *et al.* (1993) model which is an improved version of the model developed by Sekundov (1971). It has been shown in the Russian literature to solve a variety of incompressible and compressible flow problems (see Gulyaev *et al.* (1993) for selected references). Another is the model by Spalart and Allmaras (1994), which was motivated by the Baldwin-Barth model, and has become a popular model among industrial users owing to its ease of implementation and relatively inexpensive cost. Even though this one-equation level of closure is heavily based on empiricism and dimensional analysis, with characterizing flow features usually accounted for on a term by term basis using phenomenological based models, it has tended to perform well on the type of flows the models have been sensitized to, and as the recent study by Shur *et al.* (1995) has shown, can even outperform some two-equation models in separating and reattaching flows.

4 Zero- and half-equation models

To this point in the presentation of the closure models, the format has been to start from a general closure (the Reynolds stress model) and proceed to the simpler closure schemes. In this section, the two least complex closure models will be discussed. However, in the presentation of these two models, the simpler

zero-equation model will be presented first and then the more complex half-equation model. The reversal in order of presentation is because the underlying assumptions associated with the half-equation model are closely associated with the algebraic model or the zero-equation model.

Throughout the earlier parts of this chapter, the presentation of the models has been confined to the high-Reynolds-number forms in order to minimize the complications associated with the presence of solid boundaries. As will be shown shortly, the historical motivation behind the development of the algebraic model was the need to account properly for the presence of a wall in the solution of turbulent flows. This fact necessitates the introduction of some of the fundamental modelling ideas associated with the presence of solid boundaries which will also serve as an introduction to the next two sections in which the ideas behind both wall functions and near-wall closure models will be discussed.

Both the zero- and half-equation models discussed are *two-layer models*, which divide the boundary layer into an inner and outer layer, and they both use mixing-length ideas to specify the length and velocity scales that are needed. This is the basis for the derivation of the mixing-length theory and serves as the best means of presenting the key ideas of the formulation. In contrast to the previous sections where the formulations were given in a general mathematical form, the discussion in this section will be confined to simple planar flows. Extension to more complex geometries is certainly possible, but each case would have to be evaluated on its own merits to determine the appropriate measures for the relevant turbulent scales.

The zero-equation model derives its name from the fact that the eddy viscosity required in the turbulent stress-strain relationship is defined from an algebraic relationship rather than a differential one. The earliest example of such a closure is Prandtl's mixing-length theory (Prandtl, 1925). In analogy with the kinetic theory of gases, Prandtl assumed the form for the turbulent eddy viscosity in a plane shear flow with unidirectional mean flow $\bar{u}_1(x_2) = \bar{u}(y)$ and shear stress $\tau_{12} = \tau_{xy} = -\nu_t d\bar{u}/dy$. The eddy viscosity was assumed to have the form

$$\nu_t = l^2 \left| \frac{d\bar{u}}{dy} \right| \quad (42)$$

where l is the mixing length which requires specification for each flow under consideration. In a free shear flow, the mixing length would be a characteristic measure of the width of the shear layer. In a planar wall-bounded flow, the mixing length l in the near-wall region would be proportional to the distance from the wall. These relationships, though simple, give rise to significant results about the structure of turbulent flows. In the case of wall-bounded flows, the law of the wall and the structure of the outer layer of the boundary-layer flow can be deduced. Several texts and reviews in the literature provide an insightful description of the physical basis for this type of modelling. These include

Tennekes and Lumley (1972), Reynolds (1976), Speziale (1991), and Wilcox (1993).

Two of the most popular and versatile algebraic models are the Cebeci-Smith (see Cebeci and Smith, 1974) and the Baldwin-Lomax (see Baldwin and Lomax, 1978) models. These modern-day mixing-length models are two-layer eddy-viscosity models that have an inner layer eddy viscosity given by

$$\text{Cebeci-Smith:} \quad \nu_{\bar{n}} = l^2 \left(\frac{\partial \bar{u}_i}{\partial x_j} \frac{\partial \bar{u}_i}{\partial x_j} \right)^{\frac{1}{2}} \quad (43)$$

$$\text{Baldwin-Lomax:} \quad \nu_{\bar{n}} = l^2 (\bar{\omega}_i \bar{\omega}_i)^{\frac{1}{2}} \quad (44)$$

where $\omega_i = \epsilon_{ijk} \bar{u}_{k,j}$ is the vorticity vector, and an outer layer eddy viscosity given by

$$\text{Cebeci-Smith:} \quad \nu_{to} = 0.0168 \bar{u}_e \delta^* F_K(y; \delta) \quad (45)$$

$$\text{Baldwin-Lomax:} \quad \nu_{to} = 0.0168 F_{wk} F_K(y; y_m/0.3) \quad (46)$$

The mixing length is defined similarly in both models for zero-pressure-gradient flows. That is,

$$l = \kappa y \left[1 - e^{-y^+/A^+} \right] \quad (47)$$

where $\kappa = 0.41$ is the von Karman constant, A^+ is the Van Driest damping coefficient, and y^+ is the distance from the wall in wall units ($u_\tau y/\nu$). In the expressions for the outer layer eddy viscosity, δ is the boundary layer thickness, δ^* is the displacement thickness, and $\bar{u}_e$ is the edge velocity. In general, the damping coefficient A^+ can be a function of the pressure gradient, but for the present purposes it will be assumed to be constant. Throughout this subsection, attention will be focused on the form of the models for zero-pressure-gradient flows; extensions that include pressure gradient effects can be found in the references cited for the particular algebraic models. The functions F_K and F_{wk} are an intermittency and a wake function, respectively. The Klebanoff intermittency function F_K is given by

$$F_K(y; \Delta) = \left[1 + 5.5 \left(\frac{y}{\Delta} \right)^6 \right]^{-1} \quad (48)$$

and the wake function F_{wk} is given by

$$F_{wk} = \min \left(y_m F_m; y_m \bar{u}_{dif}^2 / F_m \right) \quad (49)$$

with

$$F_m = \frac{1}{\kappa} \left[\max_y (l |\bar{\omega}_i|) \right] \quad (49a)$$

Where y_m is the location where F_m occurs, Δ is the boundary layer thickness δ in the Cebeci–Smith model, and Δ is $y_m/0.3$ in the Baldwin–Lomax model. The quantity $\bar{u}_{dif}$ is the difference between the maximum and minimum total velocity in the profile. Unlike the Cebeci–Smith model, the Baldwin–Lomax model does not need to know the location of the boundary layer edge. As (49a) suggests, the Baldwin–Lomax model bases the outer layer length scale on the vorticity in the layer rather than the displacement thickness, as in the Cebeci–Smith model. Extensions and generalizations to more complex flows can be found in Wilcox (1993) and Cebeci and Smith (1974).

The next level of complexity in closure modelling, which is closely related to the algebraic models just discussed, is the half-equation model. This level of closure derives its name somewhat subjectively from the fact that an ordinary differential equation is solved rather than a partial differential equation. Nevertheless, this level of closure does generalize the algebraic model by specifying a smooth functional behaviour for the eddy viscosity across the boundary layer, and by accounting in a limited way for history effects by solving a ‘transport equation’ for the maximum shear stress. Johnson and King (1985) devised this model, and since its inception it has undergone modifications (Johnson, 1987; Johnson and Coakley, 1990) to improve its predictive capabilities for a wider class of flows, in particular, compressible flows. For the present purpose, only the simpler incompressible formulation will be outlined so not to unnecessarily complicate the explanation of the model formulation.

The Johnson–King model is also a two-layer eddy-viscosity type model; however, in this model, the eddy viscosity transitions in a prescribed functional manner from the inner layer form to the outer layer form. This functional form is given by (Johnson and King, 1985)

$$\nu_t = \nu_{io}[1 - \exp(-\nu_{ii}/\nu_{io})] \quad (50)$$

In the later form of the model (Johnson and Coakley, 1990), which was used in the solution of transonic flow problems, this functional dependency is based on a hyperbolic tangent function. The inner layer eddy viscosity is given by

$$\nu_{ii} = l^2 \frac{\sqrt{\tau_{xy}}}{\kappa y} \bigg|_m \quad (51)$$

where l is the mixing length defined in (47) with $A^+ = 15$, and the subscript m denotes maximum value. In zero-pressure-gradient two-dimensional flows where the law of the wall holds, this expression for ν_{ii} corresponds to the Cebeci–Smith inner layer eddy viscosity given in (43). The outer layer eddy viscosity is given by

$$\nu_{io} = 0.0168 \bar{u}_e \delta^* F_K(y; \delta) \sigma(x) \quad (52)$$

which is the Cebeci–Smith form (45) with the addition of the factor $\sigma(x)$ that

accounts for nonequilibrium streamwise evolution of the flow. At each streamwise station, $\sigma(x)$ is adjusted so that the relation

$$\nu_t|_m = \frac{-\tau_{xy}|_m}{\partial \bar{u} / \partial y|_m} \quad (53)$$

is satisfied. The remaining quantity that is needed is $\tau_{xy}|_m \equiv \tau_m$, and this is determined from a transport equation for the shear stress τ_{xy} . Unlike the Reynolds stress closures where the transport equation for the turbulent shear stress contains modelled pressure-strain correlations and turbulent transport terms, this turbulent shear stress equation is extracted from the turbulent kinetic energy equation (cf. equation (30)) by assuming that the shear stress anisotropy $a_{12} = -\tau_{xy}/K = 0.25$ is constant at the point of maximum shear. If the viscous diffusion effects are neglected, the evolution equation for τ_m is

$$\bar{u}_m \frac{d\tau_m}{dx} = a_{12}(\sqrt{\tau_{meq}} - \sqrt{\tau_m}) \frac{\tau_m}{L_m} - C_{dif} \frac{\tau_m^{\frac{3}{2}}}{(0.7\delta - y_m)} [1 - \sigma^{\frac{1}{2}}(x)] \quad (54)$$

where τ_{meq} is the equilibrium value ($\sigma(x) = 1$) for the shear stress, $C_{dif} = 0.5$ for $\sigma(x) \geq 1$ and zero otherwise, and L_m is the dissipation length scale given by

$$L_m = \kappa y, \quad y_m/\delta \leq 0.09/\kappa \quad (55)$$

$$L_m = 0.09\delta, \quad y_m/\delta > 0.09/\kappa \quad (56)$$

Because $\sigma(x)$ is not known *a priori* at each streamwise station, it is necessary to iterate on the equation set at each station in order to determine its value.

Once again, the reader should be reminded that the Johnson-King model that has been discussed here has evolved since its inception to a slightly different form than has been presented. The alterations are straightforward and are intended to account for compressibility effects or unique features of separated flows in order to improve its predictive capability. Nevertheless, the intent here has been to present the underlying ideas behind the formulation, with the opportunity for the interested reader to refer to the suggested references for the latest modifications and applications at this time.

C Wall functions for incompressible flows

In the previous sections in which the second-moment, two-equation and one-equation models were presented, the focus was centred on the high-Reynolds-number forms. Thus, modifications due to the presence of solid boundaries or other low-Reynolds-number effects were not presented. In the case of a solid boundary, these additional effects require modifications to the models themselves or necessitate the introduction of an intermediate layer where the turbulence quantities can be specified. In this section, the construction of such a

layer with the appropriate *wall functions* will be presented. In the next section, an alternative approach is presented where the high-Reynolds-number form of the models are themselves modified.

The wall-function approach is the most common method employed in the solution of incompressible wall-bounded flows. Although their range of applicability is formally limited to flows with constant stress layers and log layers, they have been applied to more complex flows with some success. While the formulation is somewhat standard, many authors omit the details of the formulation, especially the underlying assumptions that are used. It is instructive to review these assumptions to better ascertain the range of applicability of such methods, and to set the stage for the compressible formulation to be discussed in Section III.D.

In the constant stress layer near the solid surface, where the convective terms are small in comparison with the diffusion terms, the steady form of the streamwise component of the momentum equation (5) reduces to

$$\bar{\sigma}_{xy} - \rho\tau_{xy} = \bar{\sigma}_{xy}|_w \quad (57)$$

where in the present discussion the streamwise pressure gradient is neglected and the discussion is confined to a simple planar geometry with x and y as the streamwise and normal coordinates, respectively. In applying the wall functions along more complex surfaces, the appropriate spatial coordinate is aligned along the normal to the surface. Equation (57) simply states that the total stress $\bar{\sigma}_{tot} = \bar{\sigma}_{xy} - \rho\tau_{xy}$ in the layer is equivalent to the wall shear stress $\bar{\sigma}_w$.

The incompressible law of the wall is easily obtained from either dimensional analysis or mixing-length arguments (e.g. Tennekes and Lumley, 1972) and yields

$$\frac{\partial \bar{u}}{\partial y} = \frac{(\bar{\sigma}_w/\rho)^{1/2}}{\kappa y} \quad (58)$$

or

$$\bar{u}^+ \equiv \frac{\bar{u}}{\bar{u}_\tau} = \frac{1}{\kappa} \ln y^+ + B \quad (59)$$

where $\bar{u}_\tau = (\bar{\sigma}/\rho)^{1/2}$, $y^+ = \bar{u}_\tau y/\nu$, and (from experiments) $B \approx 5$. As shown in the discussion of the algebraic closure models, the length scale in the constant stress layer varies linearly with distance from the wall, so that the eddy viscosity ν_t is given by (31) and

$$\nu_t = \kappa y \bar{u}_\tau \quad (60)$$

Consistent with the fact that the log layer is a constant stress layer, a structural equilibrium for the turbulence is assumed and can be characterized by the fact that the Reynolds shear stress anisotropy $\tau_{xy}/K = -\sqrt{C_\mu}$ achieves an equilibrium value that is independent of the upstream conditions and

dissipation rate values. This constraint, coupled with (31) and (60), leads to the following scaling of the dissipation rate in the inner layer

$$\varepsilon = \frac{K^{\frac{3}{2}}}{C_l y} \quad (61)$$

where $C_l = \kappa/C_\mu^{\frac{3}{4}} \approx 2.5$. Note also that in the log layer $\varepsilon^+ = 1/\kappa y^+$ and $K^+ = C_\mu^{-\frac{1}{2}}$. Equation (61) is an inertial estimate of the dissipation rate and, as such, is applicable in turbulent regions close to the wall. If the wall function interface is too close to the wall, then such an estimate would cause significant errors. In flows in which separation occurs, this formulation is unacceptable because the wall shear stress would vanish. Even though the applicability of the wall function approach is suspect in such flows, the question of computability does arise. In such cases, Launder and Spalding (1974) have replaced the velocity scale $\bar{u}_\tau$ with $K^{\frac{1}{2}}$, which does not vanish for separated flows and is consistent with the $\bar{u}_\tau$ approach in constant stress flows.

It is worthwhile to outline the implementation aspects of the wall-function formulation. Although it is not possible to outline this in complete generality because of the differing numerical grid structures and geometries that can be considered, a simple framework can serve as a basis for implementation. Consider a flat-plate boundary layer flow with the computational domain spanned by a cartesian grid.

Because the mean-flow equations are evaluated at the first interior point $y_{(2)}$ away from the wall, the wall shear $\bar{\sigma}_w$ and the surface pressure $\bar{p}_w$ must be specified for these equations. The wall shear is related directly to the friction velocity which is extracted from the log-law (59). The value of the surface pressure is extracted from the normal momentum equation, with the assumption that the normal gradient of pressure at the surface is zero and that the mean pressure is constant across the area bounded by the first grid cell. This leads to the simple relation $\bar{p}_w = \bar{p}_{(2)}$.

At the same point, the turbulent kinetic energy or stresses as well as the turbulent dissipation rate need to be specified. For the two-equation formulation, application of the wall functions is straightforward and simply involves specifying both K and ε at the first interior point $y_{(2)}$. The kinetic energy is determined from $K = \bar{u}_\tau^2/\sqrt{C_\mu}$ and the dissipation rate is determined from (61). An alternative two-layer formulation has been used by Launder and Spalding (1974) and Chieng and Launder (1980) (see also Johnson and Launder, 1982), who solved the turbulent kinetic energy equation at point (2). Since this formulation explicitly accounts for the fact that the viscous sublayer is located within the first grid cell, it is necessary to treat consistently the source terms in the kinetic energy equation since the behaviour of the dependent variables in the sublayer differs significantly from that in the fully turbulent region.

For the Reynolds stress formulation, application of the wall functions is more complex since the energy redistribution between the component stresses is

unknown. This requires solving the component stress equations at the first interior point and applying appropriate boundary conditions at the wall. For the single-layer model presented here, these boundary conditions are zero convective and diffusive flux at the wall.

The turbulent dissipation rate can be specified at the first interior point based on the local equilibrium assumption given by (61), or by a more general expression extracted from the turbulent dissipation rate equation itself. Younis (1984) extracts the value of the dissipation rate at the first interior point from the dissipation rate equation (12) by assuming streamwise convection is negligible and that ϵ varies inversely with distance from the wall in the logarithmic region. In equilibrium conditions this is consistent with (61).

The main steps for implementing the wall function approach can be outlined as follows:

- (a) Set an initial guess for the wall shear stress with either $\bar{\sigma}_w = 0.3\rho K_{(2)}$ or $\bar{\sigma}_w$ from the previous iteration.
- (b) Given the velocity profile from the previous iteration (time) step, iteratively calculate $\bar{\sigma}_w$ from the implicit relationship (59).
- (c) Update the turbulent kinetic energy $K_{(2)}$ from $\bar{u}_\tau^2/\sqrt{C_\mu}$ or the turbulent stresses $\tau_{ij(2)}$ from their respective transport equations.
- (d) Update the dissipation rate $\epsilon_{(2)}$ from (61).
- (e) Return to (b) for the next iteration (time) step.

Since the various wall function distributions are developed from log-law considerations, it is necessary that the first grid point be located in this region. However, the dissipation rate is rapidly varying in this region which necessitates the use of relatively small spacing with the next interior grid point. Such issues require that the results presented in this section be used as a general guide to the basis and implementation of incompressible wall functions; however, the individual flows with their unique geometries and computational meshes may require some alterations to the above description.

D Incompressible near-wall modelling

An alternative to the wall-function method described in the previous section is to develop closure models that are capable of being integrated directly to the wall. In this section, attention is focused on modifications of the second-moment and two-equation models. Due to the relatively limited use of the one-equation models, their near-wall extensions will not be discussed in this section. The interested reader is referred to Section II.B.3 for the appropriate references. In the context of the higher order closures, the task is to account for the presence of the wall in the various closure models. Although this effort has sometimes been called low-Reynolds-number modelling, it is more properly called near-wall

modelling to distinguish it from the separate problem of transition prediction. While these near-wall models may be applicable in the transition region, their theoretical foundation is based on asymptotic consistency in the fully turbulent region. For this reason, there will be no discussion of the extension into the transition region.

The Reynolds stress formulation presents the biggest challenge to the development of near-wall closures because of the complex behaviour of the modelled correlations in the presence of the wall. Both the pressure redistribution term and the anisotropic dissipation rate, which appear in the τ_{ij} transport equation, require modification. In addition, some researchers maintain that the pressure-velocity diffusion term, which as shown in Section II.B.1 is usually neglected, also needs to be taken into account in the near-wall region. Coupled with these issues, which focus on the stress transport equations, is the behaviour of the dissipation rate equation in the vicinity of the wall. The issue of near-wall modelling is at least as complex as the issue of developing the high-Reynolds-number models. Somewhat of a reprieve is attained when only a two-equation model is used. Because the turbulent kinetic energy equation contains an isotropic dissipation rate and no pressure redistribution term, the overall task is greatly simplified. In the remainder of this section, both a near-wall second-moment closure and near-wall two-equation model will be presented in order to get a perspective of the underlying rationale behind the near-wall corrections.

Several attempts have been made at developing near-wall closure corrections to the Reynolds stress transport equations and the dissipation rate equation. For the turbulent stress transport equations, the attempts have almost exclusively focused on extending the high-Reynolds-number pressure-strain models developed by Launder and colleagues. For the dissipation rate equation, additional terms and modified coefficients have been proposed to account for the anisotropic near-wall effects and the correct limiting behaviour at the wall. Although it is not possible to detail the modifications required for the second-moment stress closures, it is beneficial to give an overview of the types of corrections made to the various modelled correlations appearing in the equations. So *et al.* (1991) have reviewed the performance of eight different near-wall Reynolds stress models. The models were tested for asymptotic consistency with the exact transport equations and against DNS and experimental data of fully developed two-dimensional channel flow. These comparisons showed that the models of Shima (1988), Launder and Shima (1989), and Lai and So (1990) yielded the best overall performance. Only a new composite model that consists of the near-wall Lai and So (1990) model and the near-wall dissipation rate equation model of So *et al.* (1991) yielded agreement over the range of variables tested. Figure 5 shows the distribution of the various turbulent correlations in the near-wall region of fully developed channel flow for the Lai and So (1990) model. The results show the level of agreement that can be expected from near-wall second-moment closures when properly calibrated.

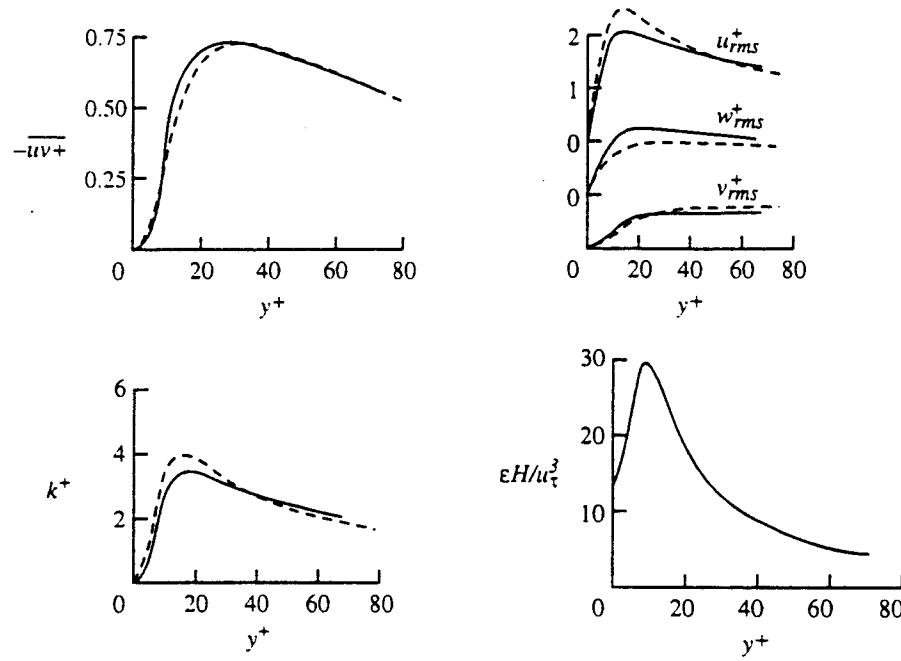


Figure 5 Near-wall distribution of turbulent correlation in fully developed channel flow (So *et al.*, 1991). Key: —, model calculations; - - -, DNS (Kim *et al.*, 1987).

It is worthwhile discussing in some detail one of the Reynolds stress near-wall models alluded to in order to gain a better perspective of the modifications required for a near-wall closure. For this purpose, the Lai and So (1990) model, which was one of the more successful models tested, will be used. Several additions are incorporated into the pressure-strain and dissipation rate equations that will be outlined. In the near-wall model, the deviatoric part of the dissipation rate is not assimilated into the pressure-strain correlation Π_{ij} but is modelled separately. The pressure-strain model is then composed of the high-Reynolds-number LRR model (15) plus a near-wall correction $\Pi_{ij}^w f_1^w$ given by

$$\Pi_{ij}^w = C_1 \varepsilon h_{ij} - \frac{\varepsilon}{K} (\tau_{ik} n_k n_j + \tau_{jk} n_k n_i) + \alpha^* (P_{ij} - \frac{1}{3} P_{kk}) \quad (62)$$

with

$$f_1^w = \exp [-(R_t/150)^2] \quad (62a)$$

where $\alpha^* = 0.45$, $R_t = K^2/\nu\epsilon$ is the usual high-Reynolds-number definition for the turbulent Reynolds number, and n_k is the unit surface normal. The form of (62) is based on an optimized balance with (15) to ensure the same asymptotic

behaviour in the near-wall region as in the exact pressure-strain correlation Π_{ij} term that appears in the τ_{ij} transport equation (8). By following the ideas of Hanjalić and Launder (1976), and Kebede *et al.* (1985) (cf. Launder and Reynolds, 1983; Launder and Tselepidakis, 1991), Lai and So (1990) used a dissipation rate model in which the anisotropy is attributed solely to the presence of the wall. The deviatoric part of the dissipation rate is then given by

$$\varepsilon_{ij}^w = \varepsilon_{ij}^w = -f_1^w \left[\frac{2}{3} \varepsilon \delta_{ij} - \frac{\varepsilon}{K} \frac{(\tau_{ij} + \tau_{ik} n_k n_j + \tau_{jk} n_k n_i + n_i n_j \tau_{kl} n_k n_l)}{(1 + 3\tau_{kl} n_k n_l / 2K)} \right] \quad (63)$$

These two modifications to the pressure-strain correlation and the tensor dissipation rate are applied to the Reynolds stress transport equation.

In the isotropic dissipation rate equation, both the production and destruction terms in (12) are modified, and an additional term is introduced to satisfy the coincidence of $\partial \varepsilon / \partial t$ and $\partial(\nu \partial^2 K / \partial x_j \partial x_j) / \partial t$ at the wall. The isotropic dissipation rate equation now becomes

$$\frac{\partial \varepsilon}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_j \varepsilon) = (1 + \sigma^w f_2^w) \bar{P}_\varepsilon - C_{\varepsilon 2} f_\varepsilon \frac{\varepsilon \varepsilon_1^w}{K} + \bar{D}_\varepsilon' + \nu \frac{\partial^2 \varepsilon}{\partial x_j \partial x_j} + \xi^w \quad (64)$$

with

$$\xi^w = f_2^w \left[\left(\frac{7}{9} C_{\varepsilon 2} - 2 \right) \frac{\varepsilon \varepsilon_1^w}{K} - \frac{1}{2} \frac{(\varepsilon_2^w)^2}{K} \right] \quad (65)$$

where $\varepsilon_1^w = \varepsilon - 2\nu(\partial\sqrt{K}/\partial x_2)^2$, $\varepsilon_2^w = \varepsilon - 2\nu K/x_2^2$, and $\sigma^w = 1 - 0.6 \exp[-Re/10^4]$. The damping function f_2^w is given by $\exp[-(Re/64)^2]$, and the modelling coefficients $C_{\varepsilon 1}$, $C_{\varepsilon 2}$, C_ε (from (12c)) take the values 1.35, 1.80, and 0.15, respectively. The variables ε_1^w and ε_2^w appear in (65) to ensure that the respective terms remain finite at the wall. The production $\bar{P}_\varepsilon$ and diffusion $\bar{D}_\varepsilon'$ terms are as given in (12a) and (12c), respectively. The form adopted here for the near-wall isotropic dissipation rate equation is similar to that adopted by Shima (1988); however, changes to the form of ε_1^w and ε_2^w have been made to increase computational efficiency. The effect of these near-wall corrections diminishes sufficiently far from the wall, so that the suitably calibrated high-Reynolds-number form of the models holds.

It would be remiss not to point out the recent work by Durbin (1993), who developed a near-wall model based on the concept of kinematic wall blocking. This concept is based on the fact that the normal component of velocity to the wall must be brought to zero inviscidly. This inviscid phenomena affects the normal velocity component at considerably greater distances from the wall than the viscous effects. Durbin (1991, 1993) accounts for this (elliptic) effect through a Poisson equation for the pressure. Both a Reynolds stress model and a simpler $K - \varepsilon - \bar{v}^2$ model (in the notation of Durbin (1991), v is the normal fluctuating velocity) are developed.

Development of a near-wall closure for a two-equation model is obviously simpler than the Reynolds stress counterpart because no pressure-strain correlation terms and anisotropic dissipation terms exist. In effect, only the requisite modifications to the isotropic dissipation rate equation are invoked, along with a suitable damping coefficient applied to the definition of the eddy viscosity (31). This modification simply alters the eddy-viscosity definition so that

$$\nu_t = C_\mu f_\mu \frac{K^2}{\varepsilon} \quad (66)$$

with the damping function f_μ given as a function of normalized distance from the wall (wall units) and also as a function of turbulent Reynolds number Re_t . With the introduction of the near-wall damping into the eddy-viscosity definition, the turbulent kinetic energy equation is then left unaltered from its high-Reynolds-number form.

Several attempts have been made to develop near-wall two-equation closures. So *et al.* (1991) have compared the performance of four different near-wall models with both DNS simulations and measurements of channel and boundary layer flows for comparison. Wilcox (1993) has compared the performance of other earlier two-equation models. This evaluation of earlier models was somewhat limited because it did not focus heavily on the performance of the turbulent quantities such as shear stress or dissipation rate. In the So *et al.* (1991) study which focused on more recent models, the models of Myong and Kasagi (1990) and Speziale *et al.* (1990) gave reasonable predictions of the kinetic energy and the shear stress in the wall region but poorly predicted the dissipation rate below x_2^+ values of 30. The two-equation $K-\omega$ model of Wilcox (1988a) was also tested, but in a more limited way because it was not formulated to be asymptotically consistent near the wall. The model proposed by So *et al.* (1991) gave good predictions in the $x_2^+ > 30$ region, and also gave the correct qualitative behaviour below the $x_2^+ = 30$ level. In the So *et al.* model, the turbulent kinetic energy equation (30) is used with the turbulent diffusion term given by (32) ($\sigma_K = 0.75$) and the eddy viscosity defined in (66) ($C_\mu = 0.096$). The damping function f_μ is the same as that used by Speziale *et al.* (1990) and given by

$$f_\mu = \left[1 + \frac{3.45}{\sqrt{Re_t}} \right] \tanh \left(\frac{x_2^+}{115} \right) \quad (67)$$

Once again, the form of the damping function f_μ is such that its effect diminishes away from the wall in regions where the high-Reynolds-number form of the models holds.

The isotropic dissipation rate equation used is given by (64) with $\sigma^w = 0$ and $\bar{D}_\varepsilon'$ given by (33) ($\sigma_\varepsilon = 1.45$). The coefficients $C_{\varepsilon 1}$ and $C_{\varepsilon 2}$ are assigned values of 1.5 and 1.83, respectively. The form of the additional term ξ^w is simplified and is

now given by

$$\xi^w = f_2^w \left[-2 \frac{\varepsilon \varepsilon_1^w}{K} + 1.5 \frac{\varepsilon_2^{w2}}{K} \right] \quad (68)$$

where f_2^w is the same as that used in (65). Additional insight into the behaviour of two-equation models can be extracted from a perturbation analysis of the equations across the boundary layer. Wilcox (1993) has analysed several two-equation models asymptotically to assess their ability to predict the correct flow behaviour in the viscous sublayer, the log layer, and the defect layer of the turbulent boundary layer.

As evident from the discussion in this section, near-wall models can and have been developed which, at the very least, give good qualitative predictions of both mean and turbulent flow quantities in simple planar wall-bounded flows. For some flow variables, the predictions are quantitatively accurate as well. The remaining issue with any of the models developed for this technologically important class of flows is their ability to predict the relevant complex engineering flows. Insufficient tests have been conducted to answer this question. However, if the models that are employed are not sufficiently mathematically consistent with the exact equation set, then the near-wall equations clearly introduce a stiffness into the overall equation set and make it difficult to solve the entire system efficiently over the entire domain.

III COMPRESSIBLE TURBULENT MODELLING

Once again, the starting point for the formulation of the compressible analogue to the Reynolds-averaged Navier–Stokes equations is the mass and momentum conservation equations, now coupled to the conservation of energy equation. The mass and momentum conservation equations are now best written in the form (cf. equations (1) and (2))

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_j) = 0 \quad (69)$$

$$\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial (\rho u_i u_j)}{\partial x_j} = - \frac{\partial p}{\partial x_i} + \frac{\partial \sigma_{ij}}{\partial x_j} \quad (70)$$

where σ_{ij} is the deviatoric part of the viscous stress tensor and is given by the relationship

$$\sigma_{ij} = 2\mu \left(S_{ij} - \frac{1}{3} S_{kk} \delta_{ij} \right) \quad (70a)$$

and p is now a thermodynamic variable. In (70a), the bulk viscosity is assumed to be small, which is the case for dilute monatomic gases.

For a perfect gas, the total energy is the sum of the internal energy and the kinetic energy, and the corresponding conservation equation is

$$\frac{\partial}{\partial t} \left[\rho \left(e + \frac{u_i u_i}{2} \right) \right] + \frac{\partial}{\partial x_j} \left[u_j \rho \left(e + \frac{u_i u_i}{2} + \frac{p}{\rho} \right) \right] = \frac{\partial}{\partial x_j} (u_i \sigma_{ij}) - \frac{\partial q_j}{\partial x_j} \quad (71)$$

where $e = c_v T$, and p is given by $\rho R T$ (R is the gas constant). The heat flux vector q_j is given by the Fourier heat conduction law $q_j = -k_T \partial T / \partial x_j$, where k_T is the thermal conductivity and c_v is the specific heat at constant volume (which is constant because the fluid is assumed to be calorically perfect). Note that the second term on the left in (71) can be written in terms of the total enthalpy $H = \rho(c_p T + u_i u_i / 2 + p / \rho)$, where c_p is the (constant) specific heat at constant pressure.

The optimal way, both mathematically and physically, to formulate both the mean conservation equations and the transport equations for any turbulence quantities is to employ Favre or mass-weighted averages (Favre, 1965, 1991). The resulting similarity between the incompressible and compressible form of the equations suggests straightforward variable density extensions of some closure models used in the incompressible formulation and also clearly identifies new correlations that are unique to the compressible formulation. For a dependent variable f , the Favre average is defined as

$$\tilde{f} = \frac{\overline{\rho f}}{\bar{\rho}}, \quad (72)$$

with the associated mean and fluctuating decomposition given by

$$f = \tilde{f} + f'' \quad (73)$$

and the time average defined in (3a). Useful guides to the properties associated with Favre averaging can be found in Rubesin and Rose (1973) and Cousteix and Aupoix (1990). If (73) is substituted into the conservation equations just discussed, the appropriate average taken, and the second moments formed, the compressible analogues to the Reynolds averaged Navier–Stokes equations and turbulent transport equations found in the incompressible formulation are obtained.

A Mean conservation equations

In the compressible formulation, additional equations for the mean conservation of mass (which replace the simpler continuity equation) and the mean conservation of (total) energy are needed. The forms of these equations and the momentum equation are intended to parallel the incompressible forms as closely as possible to highlight both the similarities and differences between the two formulations.

1 Mass and momentum

If we decompose the dependent variables according to (73) and follow the procedure outlined in the incompressible section, then the mean conservation equations for mass and momentum are given by

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_j} (\bar{\rho} \bar{u}_j) = 0 \quad (74)$$

and

$$\frac{\partial (\bar{\rho} \bar{u}_i)}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_j \bar{\rho} \bar{u}_i) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\sigma}_{ij}}{\partial x_j} - \frac{\partial (\bar{\rho} \widetilde{u_i' u_j'})}{\partial x_j} \quad (75)$$

where

$$\bar{\sigma}_{ij} = 2\bar{\mu}(\bar{S}_{ij} - \frac{1}{3}\bar{S}_{kk}\delta_{ij}) \simeq 2\bar{\mu}(\bar{S}_{ij} - \frac{1}{3}\bar{S}_{kk}\delta_{ij}) \quad (75a)$$

is the viscous stress tensor, $\bar{\mu}$ the mean molecular viscosity, and $\tau_{ij} \equiv \widetilde{u_i' u_j'}$ is the Favre-averaged correlation tensor. Equation (75a) neglects contributions from the fluctuating viscosity μ' and assumes that the Favre-averaged mean velocity and Reynolds-averaged mean velocity are approximately equal, although Ristorcelli (1993) has recently shown some evidence to the contrary.

2 Total energy

Unlike the incompressible case, the compressible formulation requires that the conservation of energy be explicitly used. In the formulation to be discussed here, a mean total energy equation with specific total energy

$$\tilde{E} = c_v \tilde{T} + \frac{\bar{u}_i \bar{u}_i}{2} + \frac{\widetilde{u_i' u_i'}}{2} \quad (76)$$

is derived. That is,

$$\frac{\partial (\bar{\rho} \tilde{E})}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_j \bar{\rho} \tilde{E}) = \frac{\partial}{\partial x_j} \bar{\Sigma}_j - \frac{\partial}{\partial x_j} (\bar{q}_j + \bar{\rho} \widetilde{E'' u_j'}) \quad (77)$$

with

$$\tilde{H} = \tilde{E} + \frac{\bar{p}}{\bar{\rho}} \quad (77a)$$

$$\bar{\Sigma}_j = \bar{\sigma}_{ij} \bar{u}_i + \bar{\sigma}_{ij} \widetilde{u_i''} + \bar{\sigma}_{ij}' \widetilde{u_i''} \quad (77b)$$

$$\bar{q}_j = -\overline{k_T T_{,j}} \simeq -\bar{k}_T \tilde{T}_{,j} \quad (77c)$$

$$\bar{\rho} \widetilde{E'' u_j'} = c_p \bar{\rho} \widetilde{u_j' T''} + \bar{\rho} \bar{u}_i \tau_{ij} + \frac{\bar{\rho} \widetilde{u_i' u_i' u_j''}}{2} \quad (77d)$$

where $\bar{k}_T$ is the mean thermal conductivity. In (77c), fluctuations in the thermal conductivity are neglected, and the Favre-averaged mean temperature and Reynolds-averaged mean temperature are taken as approximately equal. The mass flux, heat flux, and turbulent transport are additional correlations that must be modelled. An equation of state is also required for the specification of the mean pressure and for a perfect gas is given by $\bar{p} = \bar{\rho} R \bar{T}$. In terms of the total energy, $\bar{p}$ is given by the equivalent form

$$\bar{p} = (\gamma - 1) \left[\bar{\rho} \tilde{E} - \frac{1}{2} \bar{\rho} (\bar{u}^2 + \bar{v}^2 + \bar{w}^2) - \bar{\rho} K \right] \quad (78)$$

where γ is the ratio of specific heats (c_p/c_v). The presence of the turbulent kinetic energy term in the equation of state suggests a strong coupling between the mean equations and either the normal Reynolds stress components for the Reynolds stress models or the turbulent kinetic energy equation for the two-equation models.

The formulation of the mean conservation equations has yielded unknown correlations that require closure. These include the Favre-averaged correlation τ_{ij} , the turbulent heat flux $\bar{\rho} c_p u_j'' T''$, and the turbulent transport or diffusion $\bar{\rho} u_i'' u_j''$. In the incompressible formulation, a hierarchy of models for the closure of the turbulent stresses was presented as well as closures for the turbulent transport. Even at the mean equation level due to the need for an energy equation, an additional term for the turbulent heat flux has appeared, and, indirectly, for the mass flux through the equating of the Reynolds-averaged and Favre-averaged mean velocity. These additional terms, as well as the stress tensor τ_{ij} , require modelling.

B Compressible Reynolds stress and two-equation models

Even though closures for the heat and mass flux appeared in the energy conservation equation, the major impediment to future progress in treating the compressible closure problem lies with the choice of a suitable model for the turbulent stress tensor $\bar{\rho} \tau_{ij}$. The hierarchy of closure models outlined in the incompressible formulation still applies in the compressible case; however, the lower-order models as formulated do not introduce any new correlations when applied to compressible flows. Thus, in the discussion to follow, only the Reynolds stress formulation will be presented in some detail. The two-equation model will only be discussed briefly because its form is easily extracted from the full second-moment equations. The low-order models such as the zero-, half- and one-equation models will not be discussed because their extension to compressible flows is done empirically and can be found in the references that define the methods given in Sections II.B.3 and II.B.4.

Since significant similarities exist between the incompressible and compress-

ible formulations, simple variable density extensions for some models can be justified. In other cases, terms that arise solely due to the compressibility of the flow can be easily identified, and suitable models can be developed. As in the incompressible case, the starting point for the development of the Favre-averaged second-moment equation is (7) with the differential operator $\mathcal{L}$ defined from the conservation of momentum equation. The resulting Favre-averaged equation for the Reynolds stress tensor $\bar{\rho}\tau_{ij}$ is then given by

$$\frac{\partial \bar{\rho}\tau_{ij}}{\partial t} + \frac{\partial}{\partial x_k} (\bar{u}_k \bar{\rho}\tau_{ij}) = \bar{\rho}\tilde{P}_{ij} + \bar{\rho}\Pi_{ij}^d + \bar{\rho}\Pi_{ij}^{dl} + M_{ij} - \bar{\rho}\epsilon_{ij} + \frac{\partial \bar{\rho}\tilde{D}_{ijk}^t}{\partial x_k} + \frac{\partial D_{ijk}^v}{\partial x_k} \quad (79)$$

where the right-hand-side represents the rate-of-change of $\bar{\rho}\tau_{ij}$ produced by the turbulent production $\bar{\rho}\tilde{P}_{ij}$, the deviatoric part of the pressure-strain correlation Π_{ij}^d , the pressure dilatation Π_{ij}^{dl} , the mass flux variation M_{ij} , the turbulent diffusion $\tilde{D}_{ijk}^t$, the viscous diffusion D_{ijk}^v , and the turbulent dissipation rate ϵ_{ij} . These terms are given by

$$\bar{\rho}\tilde{P}_{ij} = -\bar{\rho}\tau_{ik} \frac{\partial \bar{u}_j}{\partial x_k} - \bar{\rho}\tau_{jk} \frac{\partial \bar{u}_i}{\partial x_k} \quad (79a)$$

$$\bar{\rho}\Pi_{ij}^d = \overline{p' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} - \frac{2}{3} \overline{p' \frac{\partial u'_k}{\partial x_k} \delta_{ij}} \quad (79b)$$

$$\bar{\rho}\Pi_{ij}^{dl} = \frac{2}{3} \overline{p' \frac{\partial u'_k}{\partial x_k} \delta_{ij}} \quad (79c)$$

$$M_{ij} = \overline{u'_i \left(\frac{\partial \bar{\sigma}_{jk}}{\partial x_k} - \frac{\partial \bar{p}}{\partial x_j} \right)} + \overline{u'_j \left(\frac{\partial \bar{\sigma}_{ik}}{\partial x_k} - \frac{\partial \bar{p}}{\partial x_i} \right)} \quad (79d)$$

$$\bar{\rho}\tilde{D}_{ijk}^t = -[\overline{\bar{\rho} u'_i u'_j u'_k} + \overline{p' (u'_i \delta_{jk} + u'_j \delta_{ik})}] \quad (79e)$$

$$D_{ijk}^v = \overline{(\sigma'_{ik} u'_j + \sigma'_{jk} u'_i)} \quad (79f)$$

$$\bar{\rho}\epsilon_{ij} = \overline{\sigma'_{ik} \frac{\partial u'_j}{\partial x_k}} + \overline{\sigma'_{jk} \frac{\partial u'_i}{\partial x_k}} \quad (79g)$$

The form of this compressible Reynolds stress transport equation should be compared with the incompressible form in (8).

A few comments should be made about the structuring of the pressure-strain correlation and dissipation rate terms that appear in (79). As was outlined in Section II.B.1, the deviatoric part of the dissipation rate is generally absorbed into the pressure-strain correlation to account for any anisotropic dissipation effects. The same holds for the compressible case, where the quantity

$\Pi_{ij}^d = \Pi_{ij}^d - p\epsilon_{ij}$ is modelled. Note that in the definition of the pressure-strain correlation (79b), the trace (the pressure dilatation) was subtracted out. This approach assumes that the compressibility effects associated with the pressure-strain correlation are confined to the modelling of the pressure dilatation term Π_{ij}^{dl} and that the deviatoric part Π_{ij}^d can be modelled by a simple variable mean density extension of the incompressible models presented in Section II.B.1. A recent study by Speziale *et al.* (1995) has shown, however, that in compressible homogeneous shear flow simple variable density extensions of the deviatoric part of the pressure-strain correlation leads to incorrect predictions of the Reynolds stress anisotropies. Their study indicates that such variable density extensions are only applicable in flows where the turbulent Mach number ($M_t = \sqrt{(2K/\gamma RT)}$) is about 0.3 or smaller; otherwise, deviatoric pressure-strain models must be developed that explicitly account for compressibility effects. As will be shown shortly, a similar approach can be taken for the turbulent dissipation rate.

Recall that as in the incompressible case, the deviatoric part of the tensor dissipation rate was assimilated into the pressure-strain correlation, which leaves the isotropic dissipation rate ϵ in the compressible Reynolds stress or turbulent kinetic energy formulations to be obtained from a modelled transport equation. However, unlike the incompressible case, a partitioning of the isotropic dissipation rate is generally invoked (Zeman, 1990; Sarkar *et al.*, 1991) to explicitly account for compressibility effects. The form of the partitioning is

$$\epsilon = \epsilon + \epsilon^d \quad (80)$$

where ϵ is the solenoidal (incompressible) dissipation and ϵ^d is the dilatation (compressible) dissipation. The dilatation dissipation that is unique to the compressible case will be discussed in Section III.C.4. Because the compressibility effects are represented as turbulent Mach number corrections to the incompressible (solenoidal) dissipation rate, this allows for the direct extension of the incompressible form of the dissipation rate transport equation.

This structuring is not unique, however, and an obvious alternative is to develop compressible models for the pressure-strain correlation and the dissipation rate directly. This approach is taken by El Baz and Launder (1993) in their study of compressible mixing layers. They incorporated the effects of compressibility directly into their rapid pressure-strain model through a turbulent Mach number. Because they were not only dealing with the deviatoric part of the correlation, the trace of their rapid term did not vanish in the compressible flow. Space constraints preclude the writing of the full pressure-strain model here, but its form is similar to the incompressible FLT pressure-strain correlation model developed at UMIST, which is cubic in the turbulent stress anisotropy. A similar approach is taken for the turbulent dissipation rate equation, where modifications are made to the decay term in the equation rather than explicitly partitioning the dissipation rate to account for compressibility.

The viscous diffusion D_{ijk}^v is modelled as in the incompressible case, but accounts for the variation of viscosity. Thus, the viscous diffusion term is given by (Speziale and Sarkar, 1991)

$$D_{ijk}^v \approx \bar{\mu} \left(\frac{\partial \tau_{jk}}{\partial x_i} + \frac{\partial \tau_{ki}}{\partial x_j} + \frac{\partial \tau_{ij}}{\partial x_k} \right) \quad (81)$$

Unlike the incompressible turbulent diffusion term where the pressure-velocity correlation is absorbed into the triple-correlation term and the combination modelled through gradient diffusion, a separate model for $\overline{p'u_i'}$ needs to be developed in order to be consistent with a truly compressible formulation. This need is evident by noting that $\overline{p'u_i'} \equiv \overline{p'u_i''}$ with $p' = R(\rho T'' + \rho' \tilde{T})$, which leads to the expression

$$\overline{p'u_i'} = R(\overline{\rho u_i'' T''} - \tilde{\rho} \tilde{T} \overline{u_i'}) \quad (82)$$

Equation (82) also requires models for the mass and heat flux. At this time no attempt has been made to incorporate models for either of these fluxes into the pressure-velocity diffusion term because of the inaccuracy of the models and the significance of the contribution in governing the dynamics of the flow. The correlation has simply been neglected in the modelling of the turbulent diffusion, and variable density extensions to the models given in Section II.B.1 have been used. Unfortunately, with the lack of experimental and simulation data, no *a priori* justification exists for neglecting this contribution.

However, before discussing any aspects of the dissipation rate closure, the compressible turbulent kinetic energy equation will be derived. It is easily extracted from (79) by taking the trace, which leads to

$$\frac{\partial(\bar{\rho}K)}{\partial t} + \frac{\partial}{\partial x_j} (\tilde{u}_j \bar{\rho}K) = \bar{\rho}\tilde{P} + \bar{\rho}\Pi^{dl} + \mathcal{M} + \frac{\partial}{\partial x_j} \tilde{D}_j' - \bar{\rho}\epsilon + \frac{\partial}{\partial x_j} \left(\bar{\mu} \frac{\partial K}{\partial x_j} \right) \quad (83)$$

where the right-hand-side represents the turbulent kinetic energy transport produced by the turbulent production $\bar{\rho}\tilde{P} \equiv \bar{\rho}\tilde{P}_{ii}/2$, the pressure dilatation, $\bar{\rho}\Pi^{dl} \equiv \bar{\rho}\Pi_{ii}^{dl}/2$, the mass flux variation $\mathcal{M} \equiv M_{ii}/2$, the turbulent diffusion $\tilde{D}_j' \equiv \tilde{D}_{iik}'/2$, the isotropic turbulent dissipation rate, $\epsilon \equiv \epsilon_{kk}/2$, and the viscous diffusion. In the case of the viscous diffusion, the term is not an exact contraction of the model given in (81); however, it is consistent with the incompressible and standard form given in (30). Similarly, the turbulent diffusion term is modelled as a variable density extension to the incompressible form given in (32). The other terms in the compressible turbulent kinetic energy equation that are unique to the compressible case will be discussed in the next section. Coupled with the compressible turbulent kinetic energy equation is the need for a variable density extension to the incompressible Boussinesq formula (34) for $\bar{\rho}\tau_{ij}$

$$\bar{\rho}\tau_{ij} = \frac{2}{3}\bar{\rho}K\delta_{ij} - 2\bar{\mu}_t(\tilde{S}_{ij} - \frac{1}{3}\tilde{S}_{kk}\delta_{ij}) \quad (84)$$

Note that the trace of the mean strain-rate tensor needs to be subtracted out to ensure that τ_{kk} is twice the kinetic energy. Analogous to (31), the eddy viscosity $\bar{\mu}_t$ is then given by the relation

$$\bar{\mu}_t = \bar{\rho} C_\mu \frac{K^2}{\varepsilon} \quad (85)$$

where $C_\mu = 0.09$ is a closure coefficient. As in the incompressible case, the coefficient C_μ can be made dependent on the strain-rate and rotation-rate tensors through a low-order truncation of an algebraic stress model.

The common high-Reynolds-number form of the isotropic solenoidal dissipation rate equation that is used is given by

$$\frac{\partial \bar{\rho} \varepsilon}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_j \bar{\rho} \varepsilon) = \bar{\rho} \tilde{\mathcal{P}}_\varepsilon - \frac{4}{3} \bar{\rho} \varepsilon \frac{\partial \bar{u}_k}{\partial x_k} - \bar{\rho} \tilde{\mathcal{D}}_\varepsilon + \frac{\partial}{\partial x_j} \bar{\rho} \tilde{\mathcal{D}}_{\varepsilon j} + \frac{\partial}{\partial x_j} \left(\bar{\mu} \frac{\partial \varepsilon}{\partial x_j} \right) \quad (86)$$

with

$$\tilde{\mathcal{P}}_\varepsilon = -C_{\varepsilon 1} \frac{\varepsilon}{K} \tau_{ij} \left(\frac{\partial \bar{u}_i}{\partial x_j} - \frac{1}{3} \frac{\partial \bar{u}_k}{\partial x_k} \delta_{ij} \right) \quad (86a)$$

$$\tilde{\mathcal{D}}_\varepsilon = C_{\varepsilon 2} \frac{\varepsilon^2}{K} \quad (86b)$$

where the first and third terms on the right-hand-side are the production and destruction of dissipation, respectively, and the last term is the viscous diffusion. The closure constants $C_{\varepsilon 1}$ and $C_{\varepsilon 2}$ are assumed to take on their incompressible values. The form of (86) differs slightly from the incompressible form with the addition of the mean dilatation term and the traceless production-of-dissipation term. Speziale and Sarkar (1991) have added these terms in order to account properly for the behaviour of compressed isotropic turbulence. Omission of the mean dilatation term causes the model to predict incorrectly the decrease of the integral length scale for isotropic expansion and the increase for isotropic compression.

The turbulent diffusion term is given by

$$\bar{\rho} \tilde{\mathcal{D}}_{\varepsilon j} = \bar{\rho} C_\varepsilon \left(\tau_{ij} \frac{K}{\varepsilon} \frac{\partial \varepsilon}{\partial x_i} \right) \quad (87)$$

for Reynolds stress models and by

$$\bar{\rho} \tilde{\mathcal{D}}_{\varepsilon j} = \frac{\bar{\mu}_t}{\sigma_\varepsilon} \frac{\partial \varepsilon}{\partial x_j} \quad (88)$$

for two-equation models. Both C_ε and σ_ε are also assumed to take on their incompressible values. Note that the basic form of the dissipation rate equation (86) is simply a variable density extension of the incompressible form.

Another contribution to the right-hand-side, which is also generally

neglected, arises from accounting for the variation of mean kinematic viscosity $\bar{\nu}$ in the development of the dissipation rate transport equation (Coleman and Mansour, 1991). This accounting leads to the additional term that is also proportional to the mean dilatation. For the most part, however, all these mean dilatation effects can be neglected in flows without shocks or in the absence of strong pressure gradients.

As mentioned previously, El Baz and Launder (1993) have taken an alternative approach by simply using a solenoidal dissipation rate equation that has been sensitized to compressibility effects through the turbulent Mach number. These effects were accounted for through a modification of the decay coefficient $C_{\varepsilon 2}$ rather than the solenoidal and dilatational partitioning that has been discussed. They argue that this approach is more consistent with the single-scale framework in which the dissipation equation provides a model for the energy transfer to the small scales. In their formulation, a modified coefficient $C_{\varepsilon 2c}$ was defined to account for the compressibility effects

$$C_{\varepsilon 2c} = \frac{C_{\varepsilon 2}}{1 + \beta_2 M_t^2} \quad (89)$$

where $\beta_2 = 3.2$ was obtained from calibration with the DNS of decaying isotropic turbulence.

Both Huang *et al.* (1994) and Wilcox (1992) have pointed out that the dissipation equation (86) is inappropriate in the log layer of a wall-bounded flow because it incorrectly accounts for the mean density variation near the wall. This causes the slope of the Van Driest velocity in the log-layer to be different than κ^{-1} (where κ is the Von Karman constant). However, Wilcox (1992) has shown that the $K-\omega$ formulation yields good log-law predictions in the absence of any dilatational dissipation corrections.

C Scalar flux and dilatation closure models

The closure schemes presented in the last section clearly show that the compressible formulation is complicated by the addition of second-order correlations. These include the mass flux, the turbulent heat flux, the pressure dilatation, and the compressible (dilatation) dissipation. Turbulent heat flux effects are present in the incompressible formulation but are essentially decoupled from the mean momentum field. The models to be presented and discussed reflect relatively recent contributions to the modelling of these terms. Although, this discussion may not be all inclusive, it does present the latest trends in the field. The references provided can lead the interested reader to a more complete historical record of the research in a particular topical area.

1 Mass Flux

The average fluctuating velocity $\overline{u_i''}$ is related to the mass flux $\overline{\rho' u_i'}$ through the defining relation

$$\overline{u_i''} \equiv -\frac{\overline{\rho' u_i'}}{\bar{\rho}} \quad (90)$$

Several recent attempts have been made to model this correlation, although none of the models have been extensively tested in a variety of compressible flows. Although earlier models have been proposed (e.g. see Rubesin (1990) for a brief historical survey), no dominant form has emerged. For this reason, the main attention in this survey will be on recent attempts at such modelling in conjunction with the latest advances in closure modelling and simulation data for verification.

Rubesin (1990) assumed that the density and pressure fluctuations were related through a polytropic gas law. In addition, by assuming constant specific heats to relate the fluctuating density and enthalpy, and a simple gradient transfer hypothesis between the fluctuating and mean enthalpy, he was able to relate the fluctuating velocity $\overline{u_i''}$ to moments of the turbulent Mach number

$$\overline{u_i''} \simeq \frac{1}{n-1} c_e \tau_{ij} \frac{K \bar{h}_{,j}}{\bar{h}} = \frac{\gamma-1}{n-1} c_e \frac{K}{\bar{\epsilon}} \bar{h}_{,j} \left[\frac{\tau_{ij}}{\gamma R \bar{T}} \right] \quad (91)$$

where $\bar{h}$ is the static enthalpy ($\bar{h} = \bar{H} - \bar{u}_i \bar{u}_i / 2 - \overline{u_i'' u_i''} / 2$), $c_e = 0.35$ is a closure coefficient, and n is the polytropic coefficient.

A more complicated formulation proposed by Taulbee and VanOsdol (1991) requires the solution of a modelled transport equation for the mass fluctuating velocity. In addition, because the density variance appears as a source term in the mass fluctuating velocity transport equation, an additional modelled transport equation for the density variance is derived. The general form of their mass flux equation is given by

$$\frac{D \overline{\rho' u_i'}}{Dt} = -C_{u_2} \frac{\bar{\epsilon}}{K} \overline{\rho' u_i'} + \mathcal{P}_{mf} + \mathcal{D}_{mf}' + \mathcal{D}_{mf}'' \quad (92)$$

where $C_{u_2} = 5.3$ is a model parameter determined from comparison with experimental data. The contributions to the production $\mathcal{P}_{mf}$ are from both mean velocity and mean density gradients, and the diffusion terms $\mathcal{D}_{mf}'$ and $\mathcal{D}_{mf}''$ are closed by gradient diffusion hypotheses. The first term on the right-hand-side is a modelled term that represents the relaxation of $\overline{u_i''}$. The reader is referred to Taulbee and VanOsdol (1991) for details of the formulation and the full form of the modelled transport equations. Validation tests were made on an adiabatic flat-plate boundary layer flow and a compressible free shear layer flow. In both cases, the results compared favourably with experiments.

Zeman and Coleman (1991) also derived a transport equation for the mass

flux in order to study the response of turbulence to a shock. Their transport equation consisted of a relaxation term needed to represent the decay of the turbulence after the shock, and two production-type terms based on mean density and mean velocity gradients. To lowest order, their transport equation was given by

$$\frac{D\overline{\rho'u_i'}}{Dt} \simeq -\frac{\overline{\rho'u_i'}}{\tau_a} - \tau_{ij}\bar{\rho}_{,j} - \overline{\rho'u_j'u_{i,j}} \quad (93)$$

where $\tau_a = 0.4M_t K/\varepsilon$ is an acoustic time-scale and $M_t = \sqrt{2K/\gamma R\bar{T}}$ is the turbulent Mach number with γ as the ratio of specific heats (c_p/c_v).

In the most recent attempt at developing a mass flux model, Ristorcelli (1993) has proposed an algebraic relationship extracted from an exact evolution equation for the mass fluctuating velocity that keeps only the lowest-order terms in the fluctuating density. A linear relaxation model analogous to that used by Zeman and Coleman (1991) is used to model a fluctuating divergence term. The resulting model is given by

$$\overline{u_i''} \simeq \tau [\nu_0 \delta_{ij} + \nu_1 \tau \bar{u}_{i,j} + \nu_2 \tau^2 \bar{u}_{i,k} \bar{u}_{k,j}] \tau_{jl} \frac{\bar{p}_{,l}}{\bar{\rho}} \quad (94)$$

where $\tau_a = 0.4M_t K/\varepsilon$ is an acoustic time-scale and $M_t = \sqrt{2K/\gamma R\bar{T}}$ is the turbulent Mach number with γ as the ratio of specific heats (c_p/c_v).

kinetic energy production $\bar{P}$ as defined previously. The coefficients ν_0 , ν_1 , and ν_2 are functions of the invariants of $\delta_{ij} + \bar{u}_{i,j}$. Preliminary validation tests have been made that utilize DNS data from the early stages of a compressible boundary layer flow (Dinavahi *et al.*, 1994). These tests suggest that the mass flux is important in the vicinity of a wall and that the proposed model displays the necessary qualitative features. Further validation tests are necessary that utilize DNS data of more fully developed turbulence to substantiate these preliminary results.

2 Turbulent heat flux

Unlike the incompressible case, where the mean momentum equation is decoupled from the mean energy equation with the velocity field independent of the temperature field, a direct coupling exists in the compressible case which requires a more careful specification of the turbulent heat flux modelling.

Gradient-diffusion models have been the most popular and simplest closures used for the turbulent heat fluxes, although they have not been rigorously justified in the general situation. Although several models have been derived with the assumption of a constant Prandtl number, in the presence of walls and, in particular, with non-adiabatic wall conditions, a variable Prandtl number formulation is needed. Sommer *et al.* (1993) have proposed a gradient-diffusion model that displays the correct asymptotic consistency in the near-wall region.

Their model is given by

$$\widetilde{u_i'' T''} = -\kappa_T \frac{\partial \widetilde{T}}{\partial x_i} \quad (95)$$

with turbulent heat diffusivity

$$\kappa_T = C_\lambda f_\lambda K \left[\frac{K \widetilde{T''^2}}{\varepsilon \varepsilon_T} \right]^{\frac{1}{2}} \quad (95a)$$

where $C_\lambda = 0.11$ is a model constant, f_λ is a near-wall damping function that is a function of distance from the wall and turbulent Reynolds number (see Sommer *et al.* (1992) and Sommer *et al.* (1993) for complete details of the derivation), and ε_T is the dissipation rate of the temperature variance. This variable turbulent Prandtl number formulation $Pr_t = \bar{\nu}_t / \bar{\alpha}_t$ ($\bar{\nu}_t = \bar{\mu}_t / \bar{\rho}$) required transport equations for the temperature variance and temperature variance dissipation rate (Sommer *et al.*, 1993), which were solved in conjunction with a two-equation K - ε turbulence model. Their tests on incompressible channel and pipe flows and compressible flat-plate boundary layers with adiabatic and cooled-wall boundary conditions showed that the assumption of dynamic similarity between incompressible and compressible flows is valid for high-Mach-number flows. In addition, they also verified that in highly cooled-wall, compressible boundary layer flows the Prandtl number was not constant and the assumption of dynamic similarity between momentum and heat transport was not applicable.

Another approach to determining the turbulent heat flux is to utilize a modelled transport equation for the correlation itself rather than to assume a gradient transport process at the outset and determine the turbulent heat diffusivity. As shown below, the former approach requires a knowledge of the individual turbulent stress components either through a Boussinesq approximation or a second-moment closure, and the latter approach requires the determination of the temperature variance and associated dissipation rate. Huang and Coakley (1993a) have extracted an algebraic relationship from a transport equation for the heat flux by using the assumptions associated with the development of algebraic stress models. The algebraic relation takes the form

$$-\widetilde{u_k'' T''} \frac{\partial \widetilde{u_i}}{\partial x_k} - c_{1T} \widetilde{u_i'' T''} \frac{\varepsilon}{K} + c_{2T} \widetilde{u_k'' T''} \frac{\partial \widetilde{u_i}}{\partial x_k} = \tau_{ki} \frac{\partial \widetilde{T}}{\partial x_k} \quad (96)$$

where $c_{1T} = 3$ and $c_{2T} = 0.5$ are model coefficients. This equation can be solved by direct inversion to obtain the component heat flux correlations. Huang and Coakley (1993a) also tested a simple gradient transport model proposed by Ha Minh *et al.* (1985) given by

$$\widetilde{u_i'' T''} = -c_T \frac{K}{\varepsilon} \tau_{ik} \frac{\partial \widetilde{T}}{\partial x_k} \quad (97)$$

where $c_T = 0.313$ was obtained from optimization of free shear flow predictions. Implicit in both approaches is the assumption of a variable Prandtl number. These heat flux models were used in conjunction with two Reynolds stress closure models for the solution of flat-plate flows. Tests were run over a range of Mach numbers and wall conditions, including a case of hypersonic flow with a shock-boundary layer interaction. In these cases, wall functions were used, which negated the need to incorporate near-wall corrections into the closures. The results of the study showed somewhat limited predictive capability of the combined stress closure and heat flux models tested, with little difference between the algebraic model (96) and the gradient diffusion model (97). A limited number of other, older proposals have been put forth for turbulent heat flux modelling; these are briefly discussed in the review by Cousteix and Aupoix (1990).

Finally, El Baz and Launder (1993) have used transport equations for the heat-flux, temperature variance and variance dissipation rate in conjunction with a full second-moment closure to compute the compressible mixing layer. Their pressure-temperature gradient term (analogous to the pressure-strain term in the turbulent stress transport equation) is similar in structure to their model for the pressure-strain correlation. No indication is given in the study whether such complexity is necessary for the heat flux term because the primary motivation was focused on an alternative formulation for the pressure-strain and turbulent dissipation rate models. Nevertheless, predictions of both single-stream and two-stream mixing layers were satisfactory, which indicates that the effects of pressure dilatation and dilatation dissipation are important and should be included in the closure.

3 Pressure dilatation

Because the velocity gradient tensor is not traceless in compressible flow, the structuring of the compressible stress transport equation (79) explicitly accounted for the pressure dilatation term. This correlation survives in the turbulent kinetic energy equation (83). In the Reynolds stress transport equation, this term also appears in the normal stress component equations. As mentioned previously, in this form of the equation the compressibility effects due to the pressure-strain correlation are attributed to the pressure dilatation term, and the deviatoric part can be modelled in much the same way as its incompressible counterpart with simple variable mean density extensions. With the exception of recent work of El Baz and Launder (1993), this is the approach that is generally followed when compressibility effects are explicitly taken into account.

Sarkar (1992) has proposed a model for the pressure dilatation obtained from a formal solution for homogeneous turbulence that is applicable to free shear flows. The pressure field is split into an incompressible and compressible part. The DNS data were utilized to show that only the contribution from the

incompressible pressure field to the pressure dilatation correlation was significant and needed to be modelled. The model was validated against DNS solutions of compressible homogeneous shear and isotropic turbulence; it takes the relatively simple form of

$$\Pi^{dl} = 0.30\bar{\rho}KM_t b_{mn}\tilde{u}_{mn} + 0.20\bar{\rho}\epsilon M_t^2 \quad (98)$$

Zeman (1993) has extracted a model for Π^{dl} from a transport equation for the rate of change of the pressure variance. The formulation was directed toward turbulent boundary layers in quasi-equilibrium. In the analysis, the pressure dilatation was shown to be proportional to the mass flux. The previous work of Zeman and Coleman (1991) for the mass flux was used to close the model. The resulting form was given by

$$\Pi^{dl} = 2f_\rho(M_t)\bar{a}^2\tau_{ij}\frac{K}{\epsilon}\left[\frac{\bar{\rho}_{,i}\bar{\rho}_{,j}}{\bar{\rho}}\right] \quad (99)$$

where $f_\rho(M_t) = 0.2[1 - \exp(-M_t^2/0.02)]$, and $\bar{a} = \sqrt{\gamma R\tilde{T}}$ is the sound speed. An important result of the study was the fact that the resulting model was able to preserve the Van Driest compressible law of the wall. This was attributed to the fact that the effect of the density gradient was taken into account in the pressure-strain model.

For completeness, note that Taulbee and VanOsdol (1991) proposed a composite model for the pressure dilatation and the dilatation dissipation. The resulting model required the density variance to represent turbulent-turbulent interactions and the mean dilatation to represent the influence of the mean flow. By using this approach, the remaining dissipation contribution only involved the solenoidal or incompressible contribution. The results of this study were discussed previously in the section on the mass flux. This partitioning of the turbulent dissipation rate will be discussed in the next section, and, as noted previously in the discussion of the mass flux and heat flux, the dissipation rate implied in these models is the solenoidal component.

4 Dilatation dissipation

In Section III.B, the isotropic dissipation rate was partitioned into a solenoidal part and a dilatational part. The solenoidal part was modelled through a transport equation in a manner analogous to the incompressible case. In this subsection, the closure of the dilatation dissipation is presented. In both the Zeman (1990) and Sarkar *et al.* (1991) formulations, the compressible component takes the form

$$\epsilon^d = \alpha^*\mathcal{F}(M_t)\epsilon \quad (100)$$

where α^* is a closure coefficient, and $\mathcal{F}(M_t)$ is some function of the turbulent

Mach number. In the Zeman (1990) model $\alpha^* = 0.75$, and in the Sarkar *et al.* (1991) model, $\alpha^* = 0.6$. For the function $\mathcal{F}(M_t)$, Zeman (1990) prescribes the form

$$\mathcal{F}(M_t) = 1 - \exp\{-[0.5(\gamma + 1)(M_t - M_{r0})^2/\Lambda^2]\}\mathcal{H}(M_t - M_{r0}) \quad (101)$$

where $\mathcal{H}(M_t)$ is the Heaviside function, with $M_{r0} = 0.10[2/(\gamma + 1)]^{1/2}$ and $\Lambda = 0.6$ for mixing layers, and $M_{r0} = 0.25[2/(\gamma + 1)]^{1/2}$ and $\Lambda = 0.66$ for boundary layers. Zeman's (1990) values change for the boundary-layer flows because he accounts for the change in kurtosis between the mixing-layer and boundary layer flows. Sarkar *et al.* (1991) uses the simpler form

$$\mathcal{F}(M_t) = M_t^2 \quad (102)$$

These models have been tested on simple free shear and wall-bounded flows (Sarkar *et al.* 1991; Wilcox, 1992; Zeman, 1993). In the compressible mixing layer flow, for example, this compressibility correction to the dissipation rate has proved essential in correctly predicting the spreading rate. In simple flat-plate boundary layer flows, the results are less encouraging because the log-law is adversely affected by these models, particularly in the cold-wall case. Wilcox (1992) has developed a composite dilatation dissipation model that utilizes both the Zeman and Sarkar *et al.* models for use with his $K-\omega$ formulation. The model for ε^d follows the same functional form as (100), but with $\alpha^* = 1.5$, $M_{r0} = 0.25$, and

$$\mathcal{F}(M_t) = [M_t^2 - M_{r0}^2]\mathcal{H}(M_t - M_{r0}) \quad (103)$$

If the $K-\omega$ model is used in conjunction with this dilatation dissipation model then relatively good predictions for both mixing layer and boundary layer flows are obtained. The mixing layer calculations clearly show the dramatic effect the compressibility corrections have on predicting the correct spreading rate. Figure 6 shows the unmodified $K-\omega$ model as well as the $K-\omega$ model with the Wilcox, Sarkar and Zeman compressibility corrections. Similar results are obtained from a second-moment closure formulation (Zeman, 1990; Sarkar and Lakshmanan, 1991).

As mentioned previously, Zeman (1993) has also alleviated the problem with the Van Driest scaling by accounting for density effects in the specification of his pressure dilatation model

D Wall functions for compressible flows

The discussion in the previous compressible flow sections has avoided the complicated aspects of near-wall effects. As in the incompressible case, these near-wall effects are most commonly handled through the wall function approach. The underlying assumptions used in Section II.C carry over to the

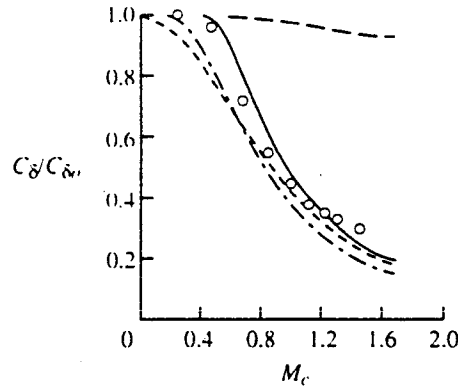


Figure 6 Comparison of computed measured spreading rate for compressible mixing layer (Wilcox, 1993). Key: ---, unmodified $K - \omega$; —, Wilcox model, - · -, Sarkar model; · · · -, Zeman model; ○, Langley curve (Kline *et al.*, 1982).

compressible case; however, the compressible formulation also includes the energy equation, which must have a temperature or heat flux specified at the wall in order to be solved. In this section, the results in Section II.C will be extended to account for this added equation. If the simple cartesian coordinate representation is used, as in the incompressible case, and, again pressure gradient effects are neglected, the momentum equation (75) and total energy equation (77) are to lowest order

$$\bar{\sigma}_{xy} - \bar{\rho}\tau_{xy} = \bar{\sigma}_{xy}|_w \quad (104)$$

$$\bar{q}_y + c_p \bar{\rho} \widetilde{v'' T''} = \bar{q}_y|_w + \bar{\sigma}_{xy} \bar{u} - \bar{\rho} \bar{u} \tau_{xy} \quad (105)$$

where v'' is the y component of the Favre-averaged fluctuating velocity. In addition, both the mass flux and turbulent transport are neglected as well. Analogous to the incompressible case, (104) simply states that the total stress $\bar{\sigma}_{tot} = \bar{\sigma}_{xy} - \bar{\rho}\tau_{xy}$ is equivalent to the wall shear stress $\bar{\sigma}_w$. Equation (105) then equates the total heat flux $\bar{q}_{tot} = \bar{q}_y + c_p \bar{\rho} \widetilde{v'' T''}$ to the wall heat flux and total stress transport:

$$\bar{q}_{tot} = \bar{q}_w + \bar{u} \bar{\sigma}_t \quad (106)$$

To obtain explicitly the compressible law of the wall, a relation between the velocity and temperature is needed. By using either dimensional analysis or mixing length arguments (Cebeci and Bradshaw, 1984), relations for the mean velocity and mean temperature gradients are found

$$\frac{\partial \bar{u}}{\partial y} = \frac{(\bar{\sigma}_w / \bar{\rho})^{\frac{1}{2}}}{\kappa y} \quad (107)$$

$$\frac{\partial \tilde{T}}{\partial y} = -Pr_t \frac{(\bar{q}_{tot}/\bar{\rho}c_p)}{(\bar{\sigma}_w/\bar{\rho})^{1/2}\kappa y} \quad (108)$$

By equating (107) and (108), the relationship

$$\frac{\bar{q}_{tot}}{c_p} \frac{\partial \tilde{u}}{\partial y} = -\frac{\bar{\sigma}_w}{Pr_t} \frac{\partial \tilde{T}}{\partial y} \quad (109)$$

is obtained, which leads to the heat flux at the wall $\bar{q}_w$ by using (106), (107), and (109):

$$\bar{q}_w = -c_p \frac{\bar{\mu}_t}{Pr_t} \frac{\partial \tilde{T}}{\partial y} - \tilde{u} \bar{\sigma}_w \quad (110)$$

where $\bar{\mu}_t$ is given by

$$\bar{\mu}_t = \bar{\rho} C_\mu \frac{K^2}{\varepsilon} \quad (111)$$

In order to obtain the Van Driest velocity from (107), it is necessary also to know the temperature distribution since the density is inversely proportional to the temperature. The temperature distribution (Cebeci and Bradshaw, 1984) in the inner layer can be obtained by integrating (109) and by using (106):

$$\tilde{T} \simeq \tilde{T}_w - \frac{Pr_t}{c_p} \left(\frac{\bar{q}_w \tilde{u}}{\bar{\sigma}_w} + \frac{\tilde{u}^2}{2} \right) \quad (112)$$

This equation is then used to determine either the wall temperature or wall heat flux. Note that (112) is only an approximate relationship because the integration limit extends directly to the wall. Because (109) is only valid in the log region of the flow, any integration to the wall through the viscous sublayer is precluded. Nevertheless, in the context of a single layer wall-function approach, there is no alternative but to perform the integration to the wall. More complex multilayer wall-function formulations (Viegas *et al.*, 1985) can be used, but these are cumbersome to apply to a Reynolds stress formulation. Equations (107) and (112) can be combined to obtain the Van Driest velocity as

$$\tilde{u}_c \simeq \left(\frac{2c_p \tilde{T}_w}{Pr_t} \right)^{1/2} \left[\arcsin \left(\frac{\bar{q}_w + \tilde{u} \bar{\sigma}_w}{\bar{\sigma}_w D} \right) - \arcsin \left(\frac{\bar{q}_w}{\bar{\sigma}_w D} \right) \right] \quad (113)$$

where

$$D = \left(\frac{\bar{q}_w^2}{\bar{\sigma}_w^2} + \frac{2c_p \tilde{T}_w}{Pr_t} \right)^{1/2} \quad (113a)$$

In addition, the Van Driest velocity satisfies the compressible law of the wall

$$\tilde{u}_c^+ \equiv \frac{\tilde{u}_c}{\tilde{u}_\tau} = \frac{1}{\kappa_c} \ln y^+ + B_c \quad (114)$$

where $\tilde{u}_\tau = \sqrt{(\bar{\sigma}_w/\bar{\rho}_w)}$ and $y^+ = \bar{\rho}_w \tilde{u}_\tau y / \bar{\mu}_w$. In the compressible formulation, both κ_c and B_c are no longer constants but are functions of various compressibility parameters in the flow. Huang *et al.* (1994) and So *et al.* (1994) (see also Wilcox, 1993) have shown that $\kappa_c = \kappa_c(M_\tau, B_q, Pr_t)$ and $B_c = B_c(M_\tau, B_q, Pr_t)$, where $B_q = \bar{q}_w / \bar{\rho}_w c_p \tilde{u}_\tau \tilde{T}_w$ is a dimensionless heat flux parameter. Although these dependencies generally hold, their effect is usually minimal in the low supersonic regime. Nevertheless, caution should be exercised whenever compressible wall functions are used because the usual log-law behaviour may not be properly represented.

The same procedure outlined in Section II.C for the wall shear stress and surface pressure is followed for the compressible case. In addition, the wall temperature or surface heat flux $\bar{q}_w$ is also specified for the mean energy equation. In addition, the turbulent kinetic energy or the turbulent stresses, and the turbulent solenoidal dissipation rate, are obtained in a manner similar to the incompressible case.

The main steps for implementing the wall functions for compressible flows can be outlined as follows:

- (a) Set an initial guess for the wall shear stress, with either $\bar{\sigma}_w = 0.3\bar{\rho}_w K_{(2)}$ or $\bar{\sigma}_w$ from the previous iteration.
- (b) *Isothermal condition:* Compute the heat flux at the wall $\bar{q}_w$ from (112) evaluated at the first interior point (2).
Adiabatic or specified $\bar{q}_w$ condition: Compute the temperature at the wall $\tilde{T}_w$ from (112) evaluated at the first interior point (2)
- (c) Given the temperature and velocity profiles from the previous iteration (time) step, calculate $\tilde{u}_{c(2)}$ from (113).
- (d) Update the wall shear stress $\bar{\sigma}_w$ from (114).
- (e) Repeat steps (b), (c), and (d) until convergence.
- (f) Update the turbulent kinetic energy $K_{(2)}$ from $\tilde{u}_\tau^2 / \sqrt{C_\mu}$ or the turbulent stresses $\tau_{ij(2)}$ from their respective transport equations.
- (g) Update the dissipation rate $\epsilon_{(2)}$ from (61).
- (h) Return to (a) for the next iteration (time) step.

As noted in the discussion of incompressible wall functions, grid spacing considerations need to be taken into account in the placement of the first grid point and the next interior one. In addition, care should be exercised when applying the compressible wall functions in the hypersonic regime. As the Mach number increases, the viscous sublayer becomes thicker. To keep the first interior point inside the turbulent region ($y^+ \approx 30$), the physical distance from the first interior point to the wall should be increased as the Mach number increases. As a result, computations of hypersonic flows with wall functions

have difficulty capturing flow separation (Huang and Coakley, 1993b) because the reverse flow in the viscous layer is not resolved.

The wall function implementation is straightforward and accurate when applied to flows that meet the conditions under which the various functions are derived. As in the incompressible case, more complex flows require more general conditions upon which to base the matching behaviour in the near-wall region with that of the high-Reynolds-number models valid away from the wall. The development of such near-wall models is the subject of the next section.

E Compressible near-wall models

As can be expected, considerably less research has concentrated on the development of compressible near-wall models than on their incompressible counterparts. In addition, the various compressible correlations discussed previously, such as the mass flux, heat flux, and pressure dilatation, have, in general, not been suitably developed for near-wall applications and have, therefore, been neglected in the near-wall models. Thus, the work to date has focused on modifying the variable density extensions of the pressure-strain correlation and/or the (solenoidal) dissipation rate equation.

Zhang *et al.* (1993b) have developed a second-moment near-wall closure model based on a variable density extension of the Lai and So (1990) model. This simply involves multiplying the near-wall forms given in (62) and (63) by the mean density $\bar{\rho}$. They invoke Morkovin's hypothesis (Morkovin, 1964) in order to eliminate the need to use explicitly any of the mass flux, pressure dilatation, or dilatation dissipation models. Unlike the Lai and So model, they include the pressure-echo term originally proposed in the Launder *et al.* (1975) study. The form used is given by

$$\Pi_{ij}^R = 2C^w \bar{\rho} K (\bar{S}_{ij} - \frac{1}{3} \bar{S}_{kk} \delta_{ij}) \left(\frac{K^{\frac{3}{2}}}{\varepsilon x_2} \right) \quad (115)$$

with

$$C^w = \begin{cases} C^w|_{in} - (5.8 \times 10^{-4}) M_{\infty} & M_{\infty} > 2.5 \\ C^w|_{in} & M_{\infty} \leq 2.5 \end{cases} \quad (115a)$$

with $C^w|_{in} = 4.14 \times 10^{-3} + 3 \times 10^{-3} Re_{\theta}$ for momentum thickness Reynolds number $Re_{\theta} \leq 5500$ and $C^w|_{in} = 0.0153$, otherwise. Their near-wall correction function ξ^w used in the dissipation rate equation also differs from the incompressible form and is given by

$$\xi^w = f_2^w \left(-2 \frac{\varepsilon \varepsilon_{w1}}{K} + 1.5 \frac{\varepsilon_{w2}^2}{K} - 0.75 C_{\varepsilon 1} \frac{\varepsilon}{K} P_{ii} \right) \quad (116)$$

where the damping function $f_2^w = \exp [-(Re_t/40)^2]$ and the modelling constants

$C_{\varepsilon 1}$, $C_{\varepsilon 2}$, and C_{ε} take the values 1.5, 1.83 and 0.1, respectively (see (64)). This model has been tested by Zhang *et al.* (1993b) on flat-plate boundary layer flows with adiabatic and cooled conditions up to a free-stream Mach number of 10. In addition to good agreement with experiments throughout the boundary layer, the model was shown to predict the Van Driest compressible law of the wall. Morrison *et al.* (1993) have also tested the model in compressible ramp flows with generally good qualitative agreement with experimental results.

Two-equation compressible models have also been derived. Zhang *et al.* (1993b) have extracted a two-equation model directly from the Reynolds stress model just discussed. In addition, Zhang *et al.* (1993a) have also derived a two-equation model by using a variable density extension of the two-equation model developed by So *et al.* (1991). These models were tested on flat-plate boundary layer flows with adiabatic- and cooled-wall conditions up to free-stream Mach numbers of 10. For the adiabatic-wall cases both the log-law and skin-friction were well predicted. For the cooled-wall conditions, the log-law slope varied from model to model, and the skin-friction predictions were much lower than the measurements.

IV NUMERICAL SOLUTION OF TURBULENT MODEL EQUATIONS

The previous sections in this chapter have presented the wide spectrum of turbulent closure models that are at the disposal of today's scientists and engineers for both incompressible and compressible flows. In this section, the focus will be on the application of the appropriate numerical techniques to the solution of the system of mean and turbulent equations outlined throughout the chapter. Although this task may seem rather extensive at first, the numerical algorithms presented in Chapters 2, 3 and 4 for laminar incompressible and compressible flows are readily extendable to the turbulent flow cases. For example, for the zero- and half-equation models, the only difference between the laminar and turbulent case is the introduction of a variable turbulent eddy viscosity into the diffusion term of the mean-flow equations.

The one-equation models of Baldwin-Barth and Spalart-Allmaras treat the eddy-viscosity equation independently of the mean flow. For the eddy-viscosity equation, they use an implicit algorithm with upwind differencing for the advection terms and central differencing for the diffusive terms. The source terms are treated implicitly; they satisfy various positivity constraints. Baldwin and Barth (1991) use an implicit factored ADI solver for the turbulent model equation, and Spalart and Allmaras (1994) use an approximate factorization scheme (see Section II.B.3) for their turbulent model equation.

Higher-order models such as the two-equation and Reynolds stress models

pose additional challenges because of the increase in the number of equations to be solved and the form of the source terms that arise from the production, destruction, and redistribution (Reynolds stress models) terms. Nevertheless, the extensions of the laminar solution techniques to the turbulent equation sets are relatively straightforward, so this section will focus on various unique aspects of these extensions and refer the reader to the previous chapters that deal with the respective incompressible and compressible solution methodologies.

A Pressure-velocity-based solution methodologies

The solution of the incompressible, constant-density turbulence equations is based on extensions of techniques for laminar incompressible equations that were outlined in Chapter 2. The most common approach for the incompressible equations is a pressure-velocity coupling, where the solution of the mean momentum equations is coupled with a Poisson equation for the pressure, which ensures a divergence-free velocity field. The solution procedure chosen for the mean momentum and pressure correction are only slightly modified to include the contribution due to either the turbulent Reynolds stresses or the turbulent kinetic energy. The basic solution procedure is then extended to solve additional transport equations by using the underlying discretization procedure adopted for the mean equations.

In the numerical solution of the incompressible Reynolds-averaged Navier-Stokes equations and turbulent transport equations presented in Section II, the equations can be written as

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} + \frac{\partial \bar{w}}{\partial z} = 0 \quad (117)$$

$$\frac{\partial \mathbf{Q}}{\partial t} + \frac{\partial (\mathbf{F}' - \mathbf{F}^v)}{\partial x} + \frac{\partial (\mathbf{G}' - \mathbf{G}^v)}{\partial y} + \frac{\partial (\mathbf{H}' - \mathbf{H}^v)}{\partial z} = \mathbf{S} \quad (118)$$

where $(\bar{u}, \bar{v}, \bar{w})$ are the respective (x, y, z) cartesian components of the mean velocity, $\mathbf{Q}$ is the vector of dependent (primitive) variables, $\mathbf{F}', \mathbf{G}', \mathbf{H}'$ are the inviscid (convective) fluxes, $\mathbf{F}^v, \mathbf{G}^v, \mathbf{H}^v$ are the viscous (diffusive) fluxes, and $\mathbf{S}$ represents the source terms due to production, destruction, and redistribution. For the present purposes, only the equation in cartesian (x, y, z) coordinates will be displayed. Extensions to generalized coordinates are the same as in the incompressible Navier-Stokes formulation of Chapter 2, and as discussed in standard numerical texts (e.g., Anderson *et al.*, 1984; Fletcher, 1988). Note that the vector formulation described here is an atypical notation for the incompressible numerical formulation. However, in keeping with the spirit of

this chapter to present a unified picture of turbulent flow modelling for both incompressible and compressible flow, it is appropriate to present the numerical formulations in as consistent and common a framework as possible. To aid in the discussion of the various solution algorithms used in solving incompressible turbulent flows, it is advantageous to partition the vector Q into mean momentum and turbulent parts given by

$$Q = \begin{Bmatrix} Q_{\text{momentum}} \\ Q_{\text{turb}} \end{Bmatrix} \quad (119)$$

The mean part is composed of the dependent variables from the momentum equation (5). The turbulent part is composed of the dependent variables from the Reynolds stress or turbulent kinetic energy equations (equation (8) or (30) respectively) and the dissipation rate equation (12). The resulting forms are given by

$$Q_{\text{momentum}} = \begin{Bmatrix} \bar{u} \\ \bar{v} \\ \bar{w} \end{Bmatrix}, \quad Q_{\text{turb}|RS} = \begin{Bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{zz} \\ \tau_{xy} \\ \tau_{xz} \\ \tau_{yz} \\ \varepsilon \end{Bmatrix}, \quad Q_{\text{turb}|K-\varepsilon} = \begin{Bmatrix} K \\ \varepsilon \end{Bmatrix} \quad (120)$$

These vector expressions, plus the continuity equation, show that the two-equation turbulence model yields six partial differential equations, and the second-moment model yields 11 partial differential equations required for the solution of an incompressible turbulent flow.

The inviscid fluxes are given by

$$F'_{\text{momentum}} = \begin{Bmatrix} \bar{u}^2 + \bar{p} \\ \bar{u}\bar{v} \\ \bar{u}\bar{w} \end{Bmatrix}, \quad F'_{\text{turb}|RS} = \begin{Bmatrix} \bar{u}\tau_{xx} \\ \bar{u}\tau_{yy} \\ \bar{u}\tau_{zz} \\ \bar{u}\tau_{xy} \\ \bar{u}\tau_{xz} \\ \bar{u}\tau_{yz} \\ \bar{u}\varepsilon \end{Bmatrix}, \quad F'_{\text{turb}|K-\varepsilon} = \begin{Bmatrix} \bar{u}K \\ \bar{u}\varepsilon \end{Bmatrix} \quad (121)$$

and the diffusive fluxes by

$$F^v_{\text{momentum}} = \begin{Bmatrix} \bar{\sigma}_{xx}/\bar{\rho} - \tau_{xx} \\ \bar{\sigma}_{xy}/\bar{\rho} - \tau_{xy} \\ \bar{\sigma}_{xz}/\bar{\rho} - \tau_{xz} \end{Bmatrix} \quad (122)$$

$$\mathbf{F}_{\text{turb}}^v|_{\text{RS}} = \begin{Bmatrix} \bar{\nu} \frac{\partial \tau_{xx}}{\partial x} + \bar{D}'_{xxx} \\ \bar{\nu} \frac{\partial \tau_{yy}}{\partial x} + \bar{D}'_{yyx} \\ \bar{\nu} \frac{\partial \tau_{zz}}{\partial x} + \bar{D}'_{zzx} \\ \bar{\nu} \frac{\partial \tau_{xy}}{\partial x} + \bar{D}'_{xyx} \\ \bar{\nu} \frac{\partial \tau_{xz}}{\partial x} + \bar{D}'_{xxz} \\ \bar{\nu} \frac{\partial \tau_{yz}}{\partial x} + \bar{D}'_{yzx} \\ \bar{\nu} \frac{\partial \epsilon}{\partial x} + \bar{D}'_{\epsilon x} \end{Bmatrix}, \quad \mathbf{F}_{\text{turb}}^v|_{K-\epsilon} = \begin{Bmatrix} \left(\bar{\nu} + \frac{\bar{\nu}_t}{\sigma_K} \right) \frac{\partial K}{\partial x} \\ \left(\bar{\nu} + \frac{\bar{\nu}_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x} \end{Bmatrix}$$

In the absence of applied body forces, no source terms associated with the mean momentum equations occur on the right-hand-side of (118), so that $\mathbf{S}_{\text{momentum}} = \mathbf{0}$; however, the turbulent equations do have source terms, and these are given by

$$\mathbf{S}_{\text{turb}}|_{\text{RS}} = \begin{Bmatrix} \bar{P}_{xx} + \Pi_{xx} - \epsilon_{xx} \\ \bar{P}_{yy} + \Pi_{yy} - \epsilon_{yy} \\ \bar{P}_{zz} + \Pi_{zz} - \epsilon_{zz} \\ \bar{P}_{xy} + \Pi_{xy} - \epsilon_{xy} \\ \bar{P}_{xz} + \Pi_{xz} - \epsilon_{xz} \\ \bar{P}_{yz} + \Pi_{yz} - \epsilon_{yz} \\ \bar{P}_\epsilon - \bar{D}_\epsilon \end{Bmatrix}, \quad \mathbf{S}_{\text{turb}}|_{K-\epsilon} = \begin{Bmatrix} \bar{P} - \epsilon \\ \bar{P}_\epsilon - \bar{D}_\epsilon \end{Bmatrix} \quad (123)$$

The casting of the governing equation set in the form presented clearly suggests that the pressure-velocity-based techniques used for the solution of the laminar incompressible flow problems are equally applicable to the turbulent problem. Additional transport equations to account for other effects such as heat transport or reacting and non-reacting gases, can be easily added to these vector forms in a straightforward manner with the same process outlined for the turbulent transport equations.

1 Pressure-velocity discretization techniques

Many popular techniques are available to discretize the governing equations, several of which are outlined in Chapter 2. One of the most popular of the early techniques used a staggered grid approach, where the different variables to be

solved were stored at alternate locations; the pressure was stored at a cell centre, and the velocities were stored on the cell faces on which they acted. The solution procedure usually used the SIMPLE (or variants thereof) procedure to couple the solution of the pressure to the solution of the velocities. This solution technique was extended to turbulent calculations with second-moment closures by Pope and Whitelaw (1976), Huang and Leschziner (1985) and Fu *et al.* (1988).

A major disadvantage of the staggered grid approach is that the different variables have different control volumes used in their solution. For a three-dimensional staggered grid, four different control volumes are required; four different sets of metrics are required for general, non-orthogonal coordinate systems. This limitation has hampered the extension of the staggered grid approach to the general, non-orthogonal coordinate systems required for complex configurations of interest to fluid dynamics engineers of today.

An alternative to the staggered grid approach is the collocated arrangement in which all solution variables are stored at the same location, usually the cell centre in a finite-volume scheme. The advantage of this arrangement is obvious for the complex grid systems because only a single control volume is used for all equations, and only a single set of grid metrics is required. The disadvantage of the collocated arrangement, historically, was the even-odd checkerboard oscillations that frequently arose in the solution but did not arise from the staggered grid arrangement. This obstacle was overcome by Rhie and Chow (1983) when they developed a collocated arrangement that included an artificial pressure dissipation to mimic the behaviour of the staggered grid. Comparisons of collocated grid with staggered grid results for laminar flows (Perić *et al.*, 1988) have shown that the collocated approach may converge faster, and has implementation advantages when non-orthogonal grids or multigrid techniques are introduced. Although no similar tests have been run for the turbulent case, the same improvements should carry over because the collocated structure is straightforwardly applied to the turbulent transport equations.

The semidiscrete second-order-accurate finite-volume form of (118) for a collocated grid arrangement is written as

$$\left[\frac{\nu}{\Delta t} (Q^{n+1} - Q^n) \right]_{i,j,k} + \left\{ [FA_x]_{i+\frac{1}{2},j,k} - [FA_x]_{i-\frac{1}{2},j,k} + [GA_y]_{i,j+\frac{1}{2},k} - [GA_y]_{i,j-\frac{1}{2},k} \right. \\ \left. + [HA_z]_{i,j,k+\frac{1}{2}} - [HA_z]_{i,j,k-\frac{1}{2}} - (\nu S)_{i,j,k} \right\} = 0 \quad (124)$$

where

$$F = F' - F^\nu, \quad G = G' - G^\nu, \quad H = H' - H^\nu \quad (124a)$$

are the fluxes defined at the interfaces of the computational cell that bounds the

cell average value $Q_{i,j,k}$; A_x , A_y , A_z represent the cell-face areas in the x , y , z directions, respectively (e.g. $A_x = \Delta y \Delta z$); and V is the cell volume ($= \Delta x \Delta y \Delta z$).

The interpolation procedure adopted in interpolating the calculated cell-centre data to the cell interface locations determines the accuracy and numerical characteristics of the scheme. The use of linear interpolation for all quantities leads to a second-order-accurate central-difference scheme that suffers from the even-odd decoupling and checkerboard oscillations. The interpolation developed by Rhie and Chow (1983) for the interface velocity in the convection term adds an additional term to the linear interpolation that is a third-order artificial pressure dissipation term. This additional term adds a fourth-order smoothing to each equation that does not affect the formal second-order accuracy of the differencing scheme. The value of the flow variables at the interfaces can be obtained from a variety of interpolation procedures including QUICK, SHARP, LODA and MUSCL. The QUICK procedure (see Chapter 2) has been commonly used for the mean momentum equations; however, QUICK can suffer from oscillations in regions of high gradients, and MUSCL (see Chapter 4) with a limiting function has been adapted to the turbulence transport equations to avoid oscillations. Both SHARP (Leonard, 1988) and LODA (Zhu and Leschziner, 1988) have also been formulated to limit oscillations, thus permitting their extension to turbulence transport equations as well (see also Lien and Leschziner, 1994).

The discretization of the diffusion terms is accomplished in an accurate and stable manner by using linear interpolation for the interface quantities, such as $\bar{v}_i$, and central differencing for the first derivative terms at the cell interfaces. This procedure has been satisfactory for laminar, algebraic models, and two-equation models because the components of τ_{ij} are evaluated with the eddy viscosity hypothesis, which results in a second-order derivative in the momentum equations. However, the Reynolds stress gradients are first derivative terms and are evaluated as such when a second-moment closure is used. This first-derivative evaluation can lead to a decoupling of the stresses from the momentum equations. Huang and Leschziner (1985), Obi *et al.* (1989) and Lien and Leschziner (1994) detailed interpolation procedures for the cell interface values of the τ_{ij} in the momentum equations. Huang and Leschziner (1985), in a staggered grid, split the stresses into an 'apparent' viscosity associated with one of the corresponding strain rates and a correction term to include the remainder of the stress term. The apparent viscosity term more strongly couples the Reynolds stress term to the velocity in the momentum equation and results in increased diagonal dominance and enhanced iterative stability. The approach of Obi *et al.* (1989) and Lien and Leschziner (1994) was to introduce a third-order artificial smoothing term based on the mean velocity similar to the artificial smoothing term introduced by Rhie and Chow (1983) for the convection terms. Additionally, the apparent viscosity term of Huang and Leschziner (1985) was used to enhance coupling and diagonal dominance.

2 Pressure-velocity solution algorithm

The solution of the system of equations just described is best understood by examining the discrete form of a representative turbulence transport equation for an element of $\mathbf{Q}_{\text{turb}} = \{Q_1, Q_2, \dots, Q_p\}^T$ and $\mathbf{S}_{\text{turb}} = \{S_1, S_2, \dots, S_p\}^T$, where $p = 2$ for two-equation models, and $p = 7$ for second-moment closures. The discrete component equation for Q_p can be written after the terms are collected as:

$$\left[\left(\frac{\nu}{\Delta t} + \alpha \right) Q_p^{n+1} \right]_{i,j,k} = \sum_m \alpha_m Q_{p_m}^{n+1} + \frac{\nu}{\Delta t} Q_p^n + \nu S^+ \quad (125)$$

where $\sum_m \alpha_m Q_{p_m}^{n+1}$ represents the convection and diffusion contributions from the neighbouring cells used in the formulation of the interface fluxes. The coefficients α , which contain metric and flow variables, are evaluated at time level n to obtain a linear relationship for the solution for flow variable Q_p . The source terms in the turbulent transport equations are split into positive and negative portions $\mathbf{S}_{\text{turb}} = \mathbf{S}_{\text{turb}}^- + \mathbf{S}_{\text{turb}}^+$. The negative part is included in the coefficient $\alpha_{i,j,k}$ to increase diagonal dominance and iterative stability. The positive portion is included as a source term to include the complete source contribution. Because the negative portion of the source term is included as $\alpha_{i,j,k} Q_p^{n+1}$, negative contributions to the source terms that do not have a factor Q_p can be included by dividing them by the previous iterate of the solution variable to arrange them in this form.

The common solution approach is based on the SIMPLE procedure of Patankar (1980) or variants such as SIMPLER or SIMPLEC (see Fletcher, 1988). The momentum equations, discretized similarly to (125), give an update procedure for each of the velocities. The discretized equations for the turbulence quantities (125) give an updated procedure for the turbulence field. Given an estimate of the pressure field, (125) can be solved for the momentum equations to give an estimate $\bar{u}^*$, $\bar{v}^*$, and $\bar{w}^*$ of the field $\bar{u}^{n+1}$, $\bar{v}^{n+1}$, and $\bar{w}^{n+1}$. In general, this intermediate value of the velocity field will not satisfy the divergence-free condition. The solution procedure then finds a correction to the pressure field $\delta\bar{p}$ that updates the pressure to the new time level $\bar{p}^{n+1} = \bar{p}^n + \delta\bar{p}$, and provides a velocity correction such that $\bar{u}^{n+1} = \bar{u}^* + \bar{u}^c$, and, similarly for $\bar{v}$ and $\bar{w}$, satisfies continuity. Complete details of this process are outlined in Chapter 2.

The complete algorithm for updating the solution of the incompressible equations from time level n to time level $n + 1$ with the SIMPLE algorithm can be summarized as follows:

- (i) Calculate $\bar{u}^*$, $\bar{v}^*$, and $\bar{w}^*$ from the discretized momentum equations with the interpolation of Lien and Leschziner (1994) for the Reynolds stress gradients from the solution field at time n .
- (ii) Obtain $\delta\bar{p}$ from the solution of the discrete Poisson equation for the pressure correction.

- (iii) Calculate the velocity field at time level $n + 1$ as $\bar{u}^{n+1} = \bar{u}^* + \bar{u}^c$ (similarly for $\bar{v}$ and $\bar{w}$).
- (iv) Calculate the pressure field at time level $n + 1$ as $\bar{p}^{n+1} = \bar{p}^n + \beta_r \delta \bar{p}$, where β_r is an underrelaxation parameter to improve stability and convergence.
- (v) Update each turbulence variable from its discrete transport equation (125).

For steady-state calculations, this procedure is repeated until each of the solution variables converges to a predetermined level. Underrelaxation is usually employed in all of the equations to improve the convergence, but with different relaxation parameters for the momentum equations, the pressure correction equation, and the turbulence transport equations (see Chapter 2 for additional details). The steps of the SIMPLE procedure outlined above are very similar to the procedures for the laminar flow case; the differences are the oscillation-free (Lien and Leschziner, 1994) inclusion of τ_{ij} in the momentum equations and the additional steps to update the transport equations for the turbulent quantities.

The pressure-velocity-based scheme presented here was developed for the incompressible constant-density flow. This pressure correction scheme can be extended to work for variable density and compressible flows, as indeed the velocity interpolation scheme for the collocated grid was developed by Rhie and Chow (1983) for compressible flows with two-equation turbulence models. Recently, the work of Lien and Leschziner (1994) reports the compressible formulation with a second-moment closure. An alternative to the pressure-velocity coupling approach is the density-velocity approach, where the compressible equations are integrated in time to a steady state.

B Density-velocity-based solution methodologies

In this section, the attention is once again focused on the numerical solution of the Reynolds stress and two-equation closure models. For the numerical solution of the compressible mean conservation and turbulent transport equations, presented in Section III, the system of equations is customarily written as

$$\frac{\partial \mathbf{Q}}{\partial t} + \frac{\partial(\mathbf{F}' - \mathbf{F}'')}{\partial x} + \frac{\partial(\mathbf{G}' - \mathbf{G}'')}{\partial y} + \frac{\partial(\mathbf{H}' - \mathbf{H}'')}{\partial z} = \mathbf{S} \quad (126)$$

where, once again, $\mathbf{Q}$ is the vector of dependent variables; $\mathbf{F}'$, $\mathbf{G}'$, $\mathbf{H}'$ are the inviscid (convective) fluxes; $\mathbf{F}''$, $\mathbf{G}''$, $\mathbf{H}''$ are the viscous (diffusive) fluxes, and $\mathbf{S}$ represents the source terms due to production, destruction and redistribution. As in the incompressible discussion, the vector $\mathbf{Q}$ can be partitioned into mean

and turbulent parts given by

$$\mathbf{Q} = \begin{Bmatrix} \mathbf{Q}_{\text{mean}} \\ \mathbf{Q}_{\text{turb}} \end{Bmatrix} \quad (127)$$

The mean part is composed of the dependent variables from the mean conservation of mass, momentum, and energy equations, (74), (75) and (77), respectively. The turbulent part is composed of the dependent variables from the Reynolds stress or turbulent kinetic energy equations, (79) or (83), respectively, and the dissipation rate equation (86). The resulting forms are given by

$$\mathbf{Q}_{\text{mean}} = \begin{Bmatrix} \bar{\rho} \\ \bar{\rho}\tilde{u} \\ \bar{\rho}\tilde{v} \\ \bar{\rho}\tilde{w} \\ \bar{\rho}\tilde{E} \end{Bmatrix}, \quad \mathbf{Q}_{\text{turb}|_{\text{RS}}} = \begin{Bmatrix} \bar{\rho}\tau_{xx} \\ \bar{\rho}\tau_{yy} \\ \bar{\rho}\tau_{zz} \\ \bar{\rho}\tau_{xy} \\ \bar{\rho}\tau_{xz} \\ \bar{\rho}\tau_{yz} \\ \bar{\rho}\varepsilon \end{Bmatrix}, \quad \mathbf{Q}_{\text{turb}|_{K-\varepsilon}} = \begin{Bmatrix} \bar{\rho}K \\ \bar{\rho}\varepsilon \end{Bmatrix} \quad (128)$$

where $(\tilde{u}, \tilde{v}, \tilde{w})$ are the respective (x, y, z) cartesian components of the mean velocity. These vector expressions show that the two-equation turbulence model yields seven partial differential equations, and the Reynolds stress formulation yields 12 partial differential equations required for the solution of a compressible turbulent flow. These numbers are compared with the six equations for the two-equation incompressible formulation, and the 11 equations for the Reynolds stress incompressible formulations.

The inviscid fluxes are given by

$$\mathbf{F}'_{\text{mean}} = \begin{Bmatrix} \bar{\rho}\tilde{u} \\ \bar{\rho}\tilde{u}^2 + \bar{p} \\ \bar{\rho}\tilde{u}\tilde{v} \\ \bar{\rho}\tilde{u}\tilde{w} \\ \bar{\rho}\tilde{u}\tilde{H} \end{Bmatrix}, \quad \mathbf{F}'_{\text{turb}|_{\text{RS}}} = \begin{Bmatrix} \tilde{u}\bar{\rho}\tau_{xx} \\ \tilde{u}\bar{\rho}\tau_{yy} \\ \tilde{u}\bar{\rho}\tau_{zz} \\ \tilde{u}\bar{\rho}\tau_{xy} \\ \tilde{u}\bar{\rho}\tau_{xz} \\ \tilde{u}\bar{\rho}\tau_{yz} \\ \tilde{u}\bar{\rho}\varepsilon \end{Bmatrix}, \quad \mathbf{F}'_{\text{turb}|_{K-\varepsilon}} = \begin{Bmatrix} \tilde{u}\bar{\rho}K \\ \tilde{u}\bar{\rho}\varepsilon \end{Bmatrix} \quad (129)$$

and the diffusive fluxes by

$$\mathbf{F}^v_{\text{mean}} = \begin{Bmatrix} 0 \\ \bar{\sigma}_{xx} - \bar{\rho}\tau_{xx} \\ \bar{\sigma}_{xy} - \bar{\rho}\tau_{xy} \\ \bar{\sigma}_{xz} - \bar{\rho}\tau_{xz} \\ \bar{\Sigma}_x - \bar{q}_x - \bar{\rho}\tilde{E}''u'' \end{Bmatrix} \quad (130)$$

$$\mathbf{F}_{\text{turb}}^v|_{\text{RS}} = \begin{Bmatrix} \bar{\mu} \frac{\partial \tau_{xx}}{\partial x} + \bar{\rho} \tilde{D}_{xxx}' \\ \bar{\mu} \frac{\partial \tau_{yy}}{\partial x} + \bar{\rho} \tilde{D}_{yyx}' \\ \bar{\mu} \frac{\partial \tau_{zz}}{\partial x} + \bar{\rho} \tilde{D}_{zzx}' \\ \bar{\mu} \frac{\partial \tau_{xy}}{\partial x} + \bar{\rho} \tilde{D}_{xyx}' \\ \bar{\mu} \frac{\partial \tau_{xz}}{\partial x} + \bar{\rho} \tilde{D}_{xzx}' \\ \bar{\mu} \frac{\partial \tau_{yz}}{\partial x} + \bar{\rho} \tilde{D}_{yzx}' \\ \bar{\mu} \frac{\partial \varepsilon}{\partial x} + \bar{\rho} \tilde{D}_{\varepsilon x}' \end{Bmatrix}, \quad \mathbf{F}_{\text{turb}}^v|_{K-\varepsilon} = \begin{Bmatrix} \left(\bar{\mu} + \frac{\bar{\mu}_t}{\sigma_K} \right) \frac{\partial K}{\partial x} \\ \left(\bar{\mu} + \frac{\bar{\mu}_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x} \end{Bmatrix}$$

As in the incompressible case, no source terms occur on the right-hand-side of (126) associated with the mean conservation equations in the absence of applied body forces, so that $\mathbf{S}_{\text{mean}} = 0$; however, the turbulent equations do have source terms; these are given by

$$\mathbf{S}_{\text{turb}}|_{\text{RS}} = \begin{Bmatrix} \bar{\rho} \tilde{P}_{xx} + \bar{\rho} \Pi_{xx}^d + \bar{\rho} \Pi_{xx}^{dl} + M_{xx} - \bar{\rho} \varepsilon_{xx} \\ \bar{\rho} \tilde{P}_{yy} + \bar{\rho} \Pi_{yy}^d + \bar{\rho} \Pi_{yy}^{dl} + M_{yy} - \bar{\rho} \varepsilon_{yy} \\ \bar{\rho} \tilde{P}_{zz} + \bar{\rho} \Pi_{zz}^d + \bar{\rho} \Pi_{zz}^{dl} + M_{zz} - \bar{\rho} \varepsilon_{zz} \\ \bar{\rho} \tilde{P}_{xy} + \bar{\rho} \Pi_{xy}^d + \bar{\rho} \Pi_{xy}^{dl} + M_{xy} - \bar{\rho} \varepsilon_{xy} \\ \bar{\rho} \tilde{P}_{xz} + \bar{\rho} \Pi_{xz}^d + \bar{\rho} \Pi_{xz}^{dl} + M_{xz} - \bar{\rho} \varepsilon_{xz} \\ \bar{\rho} \tilde{P}_{yz} + \bar{\rho} \Pi_{yz}^d + \bar{\rho} \Pi_{yz}^{dl} + M_{yz} - \bar{\rho} \varepsilon_{yz} \\ \bar{\rho} \tilde{P}_\varepsilon - \bar{\rho} \tilde{D}_\varepsilon \end{Bmatrix} \quad (131)$$

$$\mathbf{S}_{\text{turb}}|_{K-\varepsilon} = \begin{Bmatrix} \bar{\rho} \tilde{P} + \bar{\rho} \Pi^{dl} + \mathcal{M} - \bar{\rho} \varepsilon \\ \bar{\rho} \tilde{P}_\varepsilon - \bar{\rho} \tilde{D}_\varepsilon \end{Bmatrix}$$

The vector formulation just presented for the compressible equations clearly parallels the incompressible formulation presented in Section IV.A. In both cases, the discretization and solution algorithm techniques are easily extendable to the turbulent case. This extension has already been shown for incompressible flows; as will be shown next, it is also the case for compressible flows.

1 Density-velocity discretization techniques

As expected, a variety of techniques are available for the discretization of the equation set just described. In this subsection, one such technique will be described that is commonly used in the solution of the laminar equation set and that is also used in solving turbulent flow problems. Other methodologies are also available and can be extended in a similar manner.

A popular technique that has been successfully used in the solution of the mean and turbulent transport equations is based on a control volume approach. In this approach, the semidiscrete second-order-accurate finite-volume form of (126) is written as

$$\begin{aligned} \left(\nu \frac{\partial Q}{\partial t} \right)_{i,j,k} + \left\{ [FA_x]_{i+\frac{1}{2},j,k} - [FA_x]_{i-\frac{1}{2},j,k} + [GA_y]_{i,j+\frac{1}{2},k} - [GA_y]_{i,j-\frac{1}{2},k} \right. \\ \left. + [HA_z]_{i,j,k+\frac{1}{2}} - [HA_z]_{i,j,k-\frac{1}{2}} - (\nu S)_{i,j,k} \right\} = 0 \end{aligned} \quad (132)$$

where F , G , and H are the differences between the inviscid and viscous fluxes given in (124a) and are defined at the interfaces of the computational cell that bounds the cell-average value. Similarly, A_x , A_y , A_z represent the cell-face areas in the x , y , z directions, respectively, and ν is the cell volume. The definition of the numerical flux function that approximates the interface flux determines the characteristics of the numerical scheme. For compressible, transonic and supersonic flows, embedded regions with shocks are always a concern, because the high Reynolds numbers of flows away from boundaries results in very thin shock structures where the physical dissipation is quite small. Therefore, the numerical flux function chosen to evaluate the interface flux must combine good shock-capturing capabilities with good accuracy to resolve the turbulent regions.

Chapter 4 outlines several schemes designed to capture shocks and maintain accuracy in smooth regions of the flow. To illustrate the extension of the numerical methods to turbulent flows, this section outlines the extension of one of these techniques (the MUSCL scheme coupled with Roe's (1981) approximate Riemann solver) to the system of mean equations coupled with the turbulent transport equations. The interface flux for the finite-volume formulation is calculated in each of the three coordinate directions as the solution of a locally one-dimensional Riemann problem normal to the cell interface (the so-called operator splitting approach). The solution of the Riemann problem results in a shock wave, a contact discontinuity, and a rarefaction that evolves in time at the interface. The interface flux can then be determined as the flux at the left or right state incremented by the flux differences crossed from that state to the interface. The exact solution of the

Riemann problem requires an iterative procedure and is quite expensive. A cheaper alternative developed by Roe is to construct the solution to an approximate, linearized problem, where the jacobian matrix $A' = \partial F' / \partial Q$ is evaluated at an average state such that it satisfies the jump condition between the flux states at the right and left. The interface flux is written as the average of the interface flux calculated from the left state crossing negative running waves. This approach has been followed by Morrison (1990, 1992) and Huang and Coakley (1992) in solving both the two-equation and Reynolds stress equations.

The diffusive fluxes require the flow variables and derivatives of the flow variables to be evaluated at the cell interface. The flow variables required at this interface are obtained by averaging the neighbouring cell averages. The derivatives required are approximated with Gauss's divergence theorem by integrating around an auxiliary cell that is centred at the interface. This approximation is equivalent to a finite-volume representation of a second-order-accurate central-difference operator (Chakravarthy *et al.*, 1985; Swanson and Turkel, 1985), which is consistent with the elliptic nature of the diffusive fluxes.

The finite-volume form gives an integral over the control volume of the source terms. This form is approximated to second-order accuracy as the value of the cell centre times the volume of the cell. The source terms require the flow variables and the velocity derivatives at the cell centre. The velocity derivatives are calculated with Gauss's divergence theorem by integrating around the computational cell with the cell interface values of the velocity approximated as averages of neighbouring values.

Finite-difference approaches have also been used in the solution of compressible flow problems. The most popular approach is the MacCormack (1969) predictor-corrector scheme mentioned in Chapters 1 and 4, and described in detail in Peyret and Taylor (1983). Vandromme and Ha Minh (1986) have applied a two-dimensional implicit version of this scheme to both the two-equation and Reynolds stress models. The full equation set is solved fully coupled between the mean and turbulent equations. Knight (1984) (see also Narayanswami *et al.*, 1993) also uses the MacCormack method but employs a hybrid implicit-explicit approach. The implicit formulation is used in the viscous sublayer and transition wall regions in order to remove the restrictive CFL conditions necessitated by fine grid spacing in these regions. More recently, Sarkar and Lakshmanan (1991) have used an explicit MacCormack scheme to solve the coupled two-dimensional set of mean flow and turbulent Reynolds stress equations. Another finite-difference approach that has been followed is the Beam-Warming scheme (Beam and Warming, 1978; see also Peyret and Taylor, 1983). Sahu and Danberg (1986) have used a two-equation model to solve the compressible decoupled mean and turbulent equations. They restricted their study to axisymmetric flow and invoked the thin-layer approximation. In all of these approaches, the source term discretization is consistent with the spatial discretizations used on the remaining terms of the equations.

2 Density-velocity solution algorithm

The conservation and transport equations given by (126) must be integrated in time to determine the unsteady evolution or to calculate the steady-state solution. An implicit time integration is often used to integrate the equations to accelerate convergence to steady state and to avoid any stiffness problems that may occur. Although the equations can be integrated in a time accurate manner, this has not been extensively done. The limitation lies mainly with the modelled turbulent transport equations. As alluded to in the previous sections, the model equations are formed for equilibrium flows rather than for non-equilibrium flows. Although temporally evolving flows can be computed, it is not clear that such flows can be accurately predicted with the models developed to date. A common Euler implicit time integration scheme can be applied to a general discretization, either finite-volume or finite-difference, and written in delta form as

$$\left[\frac{\mathbf{I}}{\Delta t} + \delta_x \mathbf{A} + \delta_y \mathbf{B} + \delta_z \mathbf{C} - \mathbf{D} \right] \Delta \mathbf{Q} = -\mathbf{R}^n \quad (133)$$

where $\Delta \mathbf{Q} = \mathbf{Q}^{n+1} - \mathbf{Q}^n$; $\mathbf{A} = \partial \mathbf{F} / \partial \mathbf{Q}$, $\mathbf{B} = \partial \mathbf{G} / \partial \mathbf{Q}$, $\mathbf{C} = \partial \mathbf{H} / \partial \mathbf{Q}$, and $\mathbf{D} = \partial \mathbf{S} / \partial \mathbf{Q}$ are the jacobian matrices; and $\mathbf{R}^n$ is the discretized residual at time level n . In (133), δ_x , δ_y , δ_z are operators that represent the particular type of spatial differencing chosen (e.g. $[\delta_x \mathbf{B}] \Delta \mathbf{Q}$ implies $\delta_x(\mathbf{B} \Delta \mathbf{Q})$). Direct solution of (133) requires the inversion of a large banded system of block matrices at each time step to update the solution. As an alternative, Huang and Coakley (1992) have successfully used a line-by-line Gauss-Seidel approach to solve (133) for a wide variety of two-dimensional and axisymmetric flows. In three-dimensional calculations, an approximate procedure is usually used to invert the system given by (133).

The usual approach is to factor the implicit equation into simpler systems to invert. A common approach is the spatially split, approximate factorization scheme that can be written as

$$[\mathbf{I} - \Delta t \mathbf{D}] [\mathbf{I} + \Delta t \delta_x \mathbf{A}] [\mathbf{I} + \Delta t \delta_y \mathbf{B}] [\mathbf{I} + \Delta t \delta_z \mathbf{C}] \Delta \mathbf{Q} = -\Delta t \mathbf{R}^n \quad (134)$$

The first factor represents the implicit source term treatment. For a computational stencil in which the inviscid fluxes span five cell centres for a second-order upwind discretization or second-order central differencing with the artificial dissipation included, the three spatial factors require the inversion of a block pentadiagonal system (with a block size of 12 for Reynolds stress models or a block size of 7 for two-equation models) in each of the three coordinate directions. For central differencing of the diffusive fluxes, a three-point stencil results. Since the rate of convergence is relatively insensitive to the order of differencing (Walters and Thomas, 1989), a computational savings can be gained by treating the inviscid fluxes as first-order accurate in the implicit

sweeps, which reduces the inversion work because these are tridiagonal rather than pentadiagonal in form. No reduction in accuracy occurs at steady state because of the implementation in delta form. A second benefit of first-order implicit differencing of the inviscid fluxes is gained from the unconditional diagonal dominance of the first-order differencing.

The source term is a function of both the mean and turbulent solution variables and their derivatives. As such, the implicit operator for the source terms remains a large banded system as written. Solution procedures that decouple the mean flow solution from the turbulence equation integration inherently neglect the implicit contribution due to the velocity derivatives. Gorski *et al.* (1985) and Gorski (1986) in a $K-\epsilon$ two-equation adaptation solve the decoupled set. Because the turbulent equations lag behind the mean equations, the source terms require no special treatment and are handled explicitly. Huang and Coakley (1992) also decouple the solution of the turbulence equations from the mean flow equations. They split the source terms, on a term-by-term basis, into positive and negative parts with the positive source treated explicitly and the negative source treated implicitly. Sahu and Danberg (1986), who used the Beam-Warming discretizations, solved the mean and turbulent equations decoupled with the source terms treated implicitly but as part of one of the spatial factors. Fourth-order dissipation terms are added explicitly to control numerical instability.

The approach adopted by Morrison (1990, 1992) is to solve the equation set fully coupled; however, the implicit contribution due to the velocity derivatives is neglected so that the implicit source term operator is now a point implicit treatment that involves only the solution at point ij,k . The derivation of the jacobian of the source terms can be a very time-consuming project because of the complex form of the source terms, especially for near-wall models where damping functions are introduced that are functions of the solution variables. Additional approximations can be made to the implicit treatment of the source terms to reduce the work from a block-diagonal inversion and to reduce the effort required in deriving the jacobian. Morrison (1992), following the work of Vandromme and Ha Minh (1986) and Huang and Coakley (1992), replaced the jacobian matrix with a diagonal matrix that had $\min(S_p/Q_p, 0)$ for the (p,p) element. This step results in a scalar division for each equation rather than a block inversion. Additionally, the minimum function avoids the effect of the source term when it would result in an exponentially growing solution. Morrison and Gatski (1993) have showed, using an approximate factorization (AF) scheme, that it is possible to achieve convergence rates for two-equation and Reynolds stress models that match the convergence rate of a simple algebraic turbulence model (Baldwin-Lomax). The additional cost of calculating a turbulent flow-field with either a two-equation or second-moment turbulence model then corresponds only to the additional computational time required to solve the additional equations. Table 1 shows the relative cost in time per iteration per grid point for the different types of models, and

Table 1
Comparison of CPU cost for diagonalized AF scheme^a

Model	CPU (μ sec/grid point/iteration)	Relative cost	Theoretical cost
Laminar	11.57	1.00	1.0
Two-equation	20.28	1.75	1.4
Reynolds stress	33.28	2.88	2.4

^aCPU time for Navier-Stokes equations on NAS Cray C90 for $65 \times 65 \times 65$ grid (Morrison and Gatski, 1993).

demonstrates that it is possible to approach the 'theoretical' speed of 7:5 for two-equation models, and 12:5 for second-moment models.

Recent work on approximate factorization schemes with source terms (Shih and Chyu, 1991) recommends the inclusion of the source terms in the spatial factors rather than as a separate inversion to reduce factorization error. Including the source terms in the spatial factors would require either storing the source jacobian over the flow field to include it in all of the factors or recomputing it for each of the factors. The above approach has proven efficient enough that the additional storage or computational penalties are usually avoided. Additionally, the separation of the implicit source term into a separate factor facilitates the further approximations of the operators to reduce computational effort. An alternative to the approximate factorization method has been proposed by Lin *et al.* (1993). They use a preconditioned iterative method based on the Bi-CGSTAB method developed by Van der Vorst (1992) and the TFQMR method developed by Freund and Nachtigal (1991). The coupled set of mean and two-equation turbulence models is solved, and the convergent characteristics are compared with more traditional approaches.

Grasso and Speziale (1989) and Grasso and Falconi (1993), who use the finite-volume discretization approach described in the previous section, use a three-stage Runge-Kutta algorithm instead of the approximate factorization method just described. Because the equations are solved decoupled, they determine the source terms in a straightforward manner by using the known mean flow variables and the initial-stage turbulent flow variables in the computations.

In all of the techniques just discussed, a range of validity issues arise whenever the various turbulence models are used in transient or non-equilibrium turbulent flow computations. Initial transients sometimes have a significant effect on the robustness of the solution procedure. Improper initial conditions, coupled with poorly posed boundary conditions, and turbulence models that do not sufficiently account for violations of realizability constraints, can lead to regions of the flow where normal stress components and/or isotropic dissipation

rates become negative. If these effects are truly transient, then they can usually be remedied by imposing non-negative limiters on both the normal stress components (kinetic energy) and the dissipation rate. However, these techniques must be viewed with caution, and their effect on the solution must be closely monitored.

Practical constraints associated with the solution of such a complex system of equations always arise. In addition to the validity issues just mentioned, another difficulty that may arise in solving the turbulence model equations in the vicinity of shocks is often due to the source terms of the turbulence models. The production terms in the turbulent transport equations are proportional to the velocity derivative. Because of the definition of the eddy viscosity, for two-equation turbulence models, the production terms are proportional to the velocity derivatives *squared*. At a shock, the velocity derivative is becoming unbounded – a problem that is exacerbated as the inviscid scheme is improved to provide thinner shock profiles. The result is that the production term can become unbounded. An approach that has been used to minimize this problem (J. H. Morrison, private communication) is to limit the production term to a constant multiple of the destruction term. For two-equation models, in both the kinetic energy and dissipation rate equations, this is implemented as

$$\tilde{P} = \min(\tilde{P}, C\tilde{\rho}\epsilon) \quad (135)$$

where $C \gg 1$ is a constant taken to be of the $O(10^2)$. The large value of the constant C ensures that the limit is not enforced in any region where the turbulence dynamics are important. Nevertheless, if condition (135) is invoked in some dynamically important region, then the validity of the results is seriously in question due to the fact that the turbulence models are at best validated for flows where $\tilde{P}/\tilde{\rho}\epsilon$ are $O(1)$. Other such obstacles, unique to the particular flow cases that are examined and/or the numerical algorithms employed, can always arise. The difficulty lies in identifying whether the problem is a numerical or differential formulation one. A thorough knowledge of both the physical and numerical aspects of the problem is required in order to deal with the problem efficiently and effectively.

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